HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Adaptive Weight-Only Quantization for QKV Projections in Grouped Query Attention via Flip/Reflip

LƯU THIỆN VIỆT CƯỜNG
cuong.ltv225477@sis.hust.edu.vn

Program: DS&AI

Supervisor: Dr. Ngô Văn Linh

Department: DS&AI

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ABSTRACT

Large Language Models (LLMs) have achieved strong performance in many natural language processing tasks, but their deployment is limited by high computational cost and memory consumption. Quantization has been widely studied as an effective solution, especially weight-only quantization due to its efficiency and compatibility with existing inference pipelines. However, maintaining model accuracy under low-bit settings such as INT4 remains challenging. This issue is particularly critical in attention mechanisms, where the Query-Key-Value (QKV) projection layers are highly sensitive to quantization errors. Existing approaches, including AWQ and GPTQ, improve performance but still rely on fixed heuristics and do not fully exploit the statistical properties of weights and activations.

In this thesis, I propose a weight-only quantization method that incorporates the Flip/Reflip mechanism into QKV projection layers within Grouped Query Attention (GQA). This direction is chosen because GQA offers a structured trade-off between efficiency and representation, while QKV layers have a significant impact on overall model performance. The proposed method combines group-wise asymmetric INT4 quantization with dynamic outlier detection using the Kneedle algorithm. In addition, insights from the James-Stein approximation are used to guide the adjustment of quantization behavior, enabling better preservation of important weight distributions.

Later experimental results on modern LLMs on Chapter 4, show that the proposed method reduces quantization error and improves performance compared to baseline methods. Evaluation is conducted using both error-based metrics and downstream tasks=. These results demonstrate that integrating statistical modeling with structured attention design is an effective direction for improving low-bit quantization in large-scale language models.

Student

(Signature and full name)

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CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Large Language Models (LLMs) have achieved strong performance across a broad range of natural language processing tasks. Their practical deployment, however, remains constrained by the memory capacity, memory bandwidth, and computational resources required during inference. These constraints are especially important when a model must operate on limited hardware, serve many concurrent requests, or satisfy strict latency and energy budgets. Reducing the storage and data-movement cost of model parameters is therefore an important direction for making LLM inference more accessible and efficient.

Post-training quantization reduces these costs by representing pretrained model parameters with lower-precision numerical formats without requiring full model retraining. Weight-only quantization is particularly attractive because it compresses the model weights while retaining higher-precision activations, thereby limiting changes to the inference computation. Nevertheless, aggressive low-bit quantization can introduce substantial approximation error. At INT4 precision, the errors contributed by many weights may accumulate under the activations observed during inference, and the resulting aggregate projection residual can vary considerably across model components.

This thesis studies group-wise asymmetric INT4 weight-only quantization of the query, key, and value projection matrices in Transformer attention. In this setting, each weight matrix is partitioned into groups, and each group is represented by its own scale and asymmetric zero-point. The group-wise formulation can adapt to local weight distributions more effectively than a single matrix-wide quantizer, but it does not eliminate quantization error. Attention projections are a consequential target because perturbations to queries and keys alter attention scores, while perturbations to values alter the information aggregated by attention.

The notation used throughout this thesis treats an activation token x as a row vector. The corresponding projected row vectors are

$$q = xW_Q^T, \qquad k = xW_K^T, \qquad v = xW_V^T.$$

For a projection matrix W and its quantized counterpart $\widehat{W}$, the activation-weighted projection residual for a token is $x(\widehat{W} - W)^T$. Each output coordinate of this residual aggregates the signed contributions of many individual weight errors after they have been weighted by the corresponding activation values. Consequently, the isolated error of each quantized scalar is not, by itself, a complete measure of the functional projection error. Changes to W_Q additionally change the query representation and the attention scores formed with keys. These observations motivate two targeted but distinct correction procedures: Flip reduces an aggregate activation-weighted residual, whereas ReFlip evaluates disagreement at the attention-score level. The central problem is therefore to determine whether lightweight, structure-aware corrections can improve the fidelity of group-wise asymmetric INT4 QKV quantization without expanding the correction objectives into full model retraining.

1.2 Background and Research Motivation

The Transformer architecture relies on attention to determine how token representations exchange information. Standard multi-head attention assigns separate query, key, and value projections to each attention head. Multi-query attention and Grouped Query Attention (GQA) reduce key-value redundancy by allowing multiple query heads to share key and value heads. These variants can reduce memory use and improve decoding efficiency, but their sharing structure also means that an error in a shared key or value representation may influence several query heads. Quantization methods applied to GQA should therefore account for the distinct roles and shapes of the QKV projections.

Low-bit quantization must balance representational range against resolution. Asymmetric quantization can better represent groups whose distributions are not centered at zero, while group-wise quantization limits the effect of distribution differences across a large matrix. A standard round-to-nearest (RTN) decision selects the nearest representable quantization level for each weight and therefore minimizes the isolated scalar quantization error for that weight under the fixed quantizer. RTN does not, however, directly minimize the aggregate activation-weighted residual because signed scalar errors can reinforce or cancel after multiplication by the calibration activations. Static criteria for identifying consequential rounding decisions are not always appropriate because activation patterns and residual contributions can differ between matrices, layers, and groups. This thesis consequently considers adaptive identification based on the Kneedle algorithm [1], which identifies a knee in an ordered sensitivity curve. For ReFlip, the detected knee provides a data-dependent boundary between extremely sensitive query dimensions that are protected and a moderate post-knee region considered for correction.

The proposed framework distinguishes two correction stages. *Flip* changes selected RTN rounding decisions after group-wise asymmetric INT4 quantization in order to reduce the aggregate activation-weighted projection residual measured on calibration data. Because RTN already minimizes each isolated scalar error, a flipped decision may move an individual quantized weight farther from its original full-precision value. Flip accepts such an increase only when the changed signed contribution helps reduce the combined residual produced by the relevant weights under the observed activations. It therefore targets cancellation and accumulation in the aggregate residual rather than seeking uniformly smaller individual weight errors. Flip preserves the INT4 representation and the broader quantization configuration, and it does not use the attention-score objective described below.

ReFlip is a separate, subsequent correction stage. ReFlip updates only the query projection matrix W_Q; it does not update W_K or W_V. The update is guided by a scalar attention-score loss that measures disagreement between attention scores produced by the reference and quantized projections on calibration data. By using a scalar loss at the score level, ReFlip targets the functional effect of query-projection error without introducing a full hidden-state reconstruction objective. The method is also informed by the general motivation of statistical shrinkage: in high-dimensional estimation, carefully regularized adjustments can reduce aggregate error relative to unregularized estimates [2]. This connection is used as design motivation rather than as a claim that the proposed correction is itself a James–Stein estimator.

Together, Flip and ReFlip address different aspects of the same problem. Flip changes selected rounding decisions across QKV projections to reduce an aggregate activation-weighted projection residual, whereas ReFlip focuses narrowly on W_Q and uses attention-score disagreement as its signal. Keeping the two stages conceptually separate is important for evaluating their individual effects and for avoiding claims that improvements from one stage necessarily imply improvements from the other.

1.3 Research Questions and Objectives

The overall objective of this thesis is to investigate whether targeted corrections can improve the reliability of group-wise asymmetric INT4 weight-only quantization for QKV projections, particularly in attention modules using GQA. The study is guided by the following research questions:

1. How is quantization error distributed across QKV projection matrices and their weight groups under group-wise asymmetric INT4 weight-only

quantization?

2. Can adaptively selected Flip decisions reduce the aggregate activation-weighted QKV projection residual, even when some flipped decisions increase their individual scalar weight errors, without changing the prescribed INT4 weight-only setting?

3. Can ReFlip, defined as an update to W_Q alone using a scalar attention-score loss, improve attention-score fidelity beyond the Flip stage?

4. To what extent do changes in weight-space error and attention-score error correspond to changes in downstream language-model evaluation?

To answer these questions, the thesis develops a controlled framework that first establishes the INT4 quantization setting, then evaluates Flip and ReFlip as distinct interventions. The analysis distinguishes isolated scalar quantization error, aggregate activation-weighted projection residual, attention-score error, and downstream evaluation. This distinction is necessary because RTN can minimize each isolated scalar error while leaving a larger combined projection residual, and because a reduction in that residual does not by itself guarantee a meaningful improvement in model quality.

1.4 Scope and Boundaries

This work is limited to post-training, group-wise asymmetric INT4 weight-only quantization. Activations remain outside the low-bit quantization scope, and the thesis does not propose quantization-aware training or full model retraining. The primary objects of study are the W_Q, W_K, and W_V projection matrices in Transformer attention, with particular attention to GQA structure. Other model components may still affect end-to-end behavior, but they are not the central correction targets considered here.

Flip and ReFlip are deliberately bounded procedures. Flip changes only selected RTN rounding decisions within the quantized QKV projections and selects those changes according to their effect on the aggregate activation-weighted projection residual over calibration data. It may worsen individual scalar weight errors and is not intended to minimize a weight-wise error metric. ReFlip updates only W_Q and is optimized using a scalar attention-score loss on calibration data; it does not update W_K, W_V, or the remaining model parameters. The thesis evaluates these procedures empirically and does not claim universal improvement across all architectures, tasks, datasets, group sizes, activation distributions, or calibration choices. Likewise, adaptive selection and shrinkage-inspired reasoning are treated as practical design principles whose usefulness must be established by the reported experiments.

1.5 Contributions

Within the stated scope, this thesis makes the following contributions:

1. It formulates a targeted study of QKV projections under group-wise asymmetric INT4 weight-only quantization, with explicit consideration of the sharing structure present in GQA.
2. It defines Flip as a selective change to RTN rounding decisions that may increase individual scalar weight errors while reducing the aggregate activation-weighted QKV projection residual, and it distinguishes this objective from ReFlip.
3. It defines ReFlip as a query-only correction that updates W_Q using a scalar attention-score loss while leaving W_K, W_V, and other model parameters unchanged.
4. It incorporates Kneedle-based partitioning of query-dimension sensitivities so that ReFlip protects the extreme pre-knee region and evaluates candidates from a moderate post-knee region, and it evaluates the resulting corrections using both quantization-fidelity measures and downstream language-model evaluation.
5. It provides an empirical assessment of when the proposed corrections are helpful and where their benefits are limited, explicitly distinguishing RTN's isolated scalar-error objective from Flip's aggregate activation-weighted residual objective rather than assuming that either objective necessarily produces better end-to-end performance.

These contributions are intended as a focused investigation of attention-projection correction under a specific low-bit setting. They do not establish that QKV projections are always the dominant source of quantization degradation or that the proposed procedures replace broader post-training quantization methods.

1.6 Organization of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 reviews Transformer attention, GQA, and relevant approaches to neural network and LLM quantization. Chapter 3 presents the proposed methodology, including group-wise asymmetric INT4 weight-only quantization, adaptive selection, Flip, and the query-only ReFlip objective. Chapter 4 reports the experimental evaluation, including ablation studies, quantization fidelity analysis, and downstream results.

Chapter 5 summarizes the findings and limitations and identifies directions for future research.

CHAPTER 2. LITERATURE REVIEW

This chapter establishes the conceptual and empirical context for low-bit quantization of attention projection weights in large language models. The discussion first defines the scope of the thesis and then distinguishes the principal quantization taxonomies that are often conflated in the literature. It subsequently reviews representative post-training quantization methods, develops the mathematical background of attention and Grouped Query Attention (GQA), and analyzes how projection-weight perturbations enter attention scores and outputs. The final sections separate weight outliers, activation outliers, and functional salience; clarify the limited roles of Kneedle and James–Stein reasoning; and identify the research gap addressed by the thesis. Throughout the chapter, claims about sensitivity and error propagation are qualified by the relevant operating conditions. This distinction is important because structural reuse creates dependencies, but does not by itself prove that errors will be amplified in every input, head, layer, or task.

2.1 Scope of Research

This thesis studies post-training quantization of the query, key, and value projection weights in Transformer-based large language models, with particular attention to models that use GQA. The target setting is group-wise asymmetric INT4 weight-only quantization. The original pretrained model is not retrained or fine-tuned. Activations remain in higher precision, and the study does not attempt to optimize every layer or every inference-system component. This restricted scope makes it possible to examine how the statistical structure of QKV weights and the computational structure of attention jointly affect quantization error.

The models motivating the study include contemporary Transformer families such as LLaMA [3] and Qwen [4]. Their parameter counts make memory capacity and memory bandwidth important deployment constraints. Weight-only quantization directly reduces the storage and transfer cost of model parameters while avoiding the additional dynamic-range difficulties introduced by quantizing activations. The resulting inference process still uses higher-precision token representations, but each linear projection is formed from an approximation of its original weight matrix. Consequently, the quality of the projection depends on both the quantized weights and the activations that multiply them.

The thesis quantizer is consistently defined as *group-wise asymmetric weight-only quantization*. For a projection matrix, each row corresponds to an output channel and each column corresponds to an input channel. Contiguous weights along the input-channel dimension are partitioned into numerical quantization groups. Every such group receives its own scale and zero-point. These numerical groups are unrelated to GQA groups. A GQA group is an architectural association in which several query heads share a key head and a value head; a quantization group is merely a local partition of weights represented by common affine parameters. Maintaining this distinction prevents ambiguity when discussing group size, error sharing, or head structure.

The study also considers a lightweight post-quantization refinement referred to as *bit-level adjustment*. In this thesis, the term has a precise operational meaning: a selected stored integer weight value is changed to an adjacent integer code, and therefore to an adjacent representable quantization level, while retaining the group's scale, zero-point, and assigned bit width. Bit-level adjustment is not a literal flip of a particular binary digit. A literal binary-digit flip can move an integer by a large and representation-dependent distance, whereas the intended operation changes the quantized value by one neighboring level. The term is preserved because it emphasizes discrete low-bit refinement, but every subsequent use refers to adjacent integer-code adjustment.

Calibration data may be used to characterize the effects of quantized weights under representative inputs and to evaluate candidate adjustments. Such use does not constitute training: no gradient-based optimization of the original model parameters is required. The study also excludes hardware-specific kernel design and activation or KV-cache quantization. These topics are relevant to complete deployment systems, but separating them from projection-weight quantization keeps the evidence boundaries clear. In particular, results about quantized KV caches can motivate attention-aware analysis, yet they do not directly establish how QKV projection weights behave under group-wise asymmetric weight-only quantization.

The resulting research question is deliberately narrow. Given low-bit group-wise asymmetric representations of QKV projection weights, can selective adjacent-level adjustment reduce consequential residual errors while respecting the structure of attention and GQA? Answering this question requires more than reporting weight reconstruction error. It requires a taxonomy of quantization choices, a perturbation model that connects weight error to attention computations, and careful interpretation of statistical heuristics. These foundations are developed in the remainder of the chapter.

2.2 Quantization Concepts and Taxonomy

Quantization maps values from a high-precision domain to a finite representable set. For neural-network weights, this mapping reduces memory consumption and can reduce data-transfer cost, but it introduces approximation error. The literature classifies quantizers along several independent dimensions, including uniform versus non-uniform level spacing, symmetric versus asymmetric ranges, tensor granularity, weight-only versus weight-and-activation quantization, and post-training versus training-aware procedures [5], [6]. A precise account must keep these dimensions separate because a quantizer can, for example, be uniform, asymmetric, group-wise, weight-only, and post-training at the same time.

Uniform and Non-uniform Levels

A uniform quantizer uses a constant step size between adjacent dequantized values. If the step size is $s > 0$, moving from one integer code to its neighbor changes the dequantized value by exactly s, except where clipping prevents movement beyond the representable endpoints. Uniformity refers only to spacing. It does not require the range to be centered at zero, nor does it require all weights in a tensor to share one scale.

A non-uniform quantizer places representable values at unequal intervals. Such placement can allocate more levels to dense regions of a distribution and fewer levels to sparse regions, potentially improving representation for a specified distortion measure. The trade-off is that irregular levels can complicate representation and arithmetic. The thesis does not use a non-uniform quantizer. Its asymmetric affine quantization remains uniformly spaced within every quantization group.

Symmetric and Asymmetric Affine Quantization

Symmetry concerns the location of the represented range relative to zero. A symmetric quantizer ordinarily uses a range centered at or approximately centered at zero. This choice can simplify some arithmetic and is appropriate when positive and negative magnitudes are balanced. An asymmetric affine quantizer introduces a zero-point that shifts the represented interval. The shift can use the available integer range more effectively when a local weight group is skewed or has unequal positive and negative extrema.

Let a quantization group contain real-valued weights w_i, and let the available integer interval be $[c_{\min}, c_{\max}]$. With scale $s > 0$ and zero-point z, affine quantization and dequantization can be written as

$$c_i = \mathrm{clip}\left(\mathrm{round}\left(\frac{w_i}{s}\right) + z,\ c_{\min}, c_{\max}\right), \qquad \hat{w}_i = s(c_i - z). \tag{2.1}$$

The representable dequantized values are separated by the constant distance s. Thus, asymmetric affine quantization is still uniform quantization. Its asymmetry shifts the represented range through z; it does not create unequal spacing. For INT4, at most $2^4 = 16$ integer codes are available per group, so the choice of range and the treatment of extreme weights strongly influence the resolution assigned to the remaining weights.

The quantization error for an individual weight is $\Delta w_i = \hat{w}_i - w_i$. Before clipping, nearest rounding bounds its magnitude by approximately $s/2$. Clipping can produce a larger error because a value outside the selected range is mapped to an endpoint. This elementary distinction is important when interpreting candidate refinement: adjacent-level adjustment changes $\hat{w}_i$ by s, whereas changing the scale or clipping range changes the representable set for every weight in the group.

Granularity: Tensor, Output Channel, and Quantization Group

Quantization granularity identifies which weights share affine parameters. Per-tensor quantization assigns one scale and, where applicable, one zero-point to an entire tensor. It is economical but forces heterogeneous regions to share one range. Per-channel quantization, as a taxonomy comparator, assigns separate parameters to complete channels. It can adapt to differences across channels but still treats every weight within one channel as a single range.

The thesis instead uses group-wise quantization. Within each output-channel row, contiguous columns along the input-channel dimension are divided into quantization groups. Each group has an independent scale and zero-point. This arrangement provides finer local adaptation than per-tensor or complete-channel granularity while preserving a regular partition. The phrase “group-wise” always refers to this numerical partition unless “GQA group” is explicitly stated.

The distinction between input and output channels follows the orientation of the projection matrices used throughout this chapter. For one token representation $x \in \mathbb{R}^{d_{\text{in}}}$ and a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the projection is $xW^\top$. A row of W produces one output-channel component; a column multiplies one input feature. Grouping along the input-channel dimension therefore divides each row into contiguous sets of contributing input features. It does not combine multiple attention heads merely because those heads belong to the same GQA group.

Weight-only and Post-training Quantization

Weight-only quantization stores weights at reduced precision while leaving runtime activations in higher precision. The approximation error arises from the quantized weights, but its realized effect depends on the activations that multiply those weights. For a linear projection $y = xW^\top$ and quantized weights $\hat{W} = W + \Delta W$,

$$\hat{y} - y = x\Delta W^\top. \tag{2.2}$$

The same weight perturbation can therefore have different output effects for different inputs. A large-magnitude weight error aligned with rarely active input directions may have limited realized influence, whereas a smaller error aligned with frequently large or task-relevant activations may matter more. This relationship motivates the use of representative calibration inputs without changing the definition of the method as weight-only quantization.

Post-training quantization operates on a pretrained model without repeating the original training process. It can use weight statistics, calibration activations, local reconstruction objectives, or approximate second-order information. Quantization-aware training instead exposes model optimization to quantization effects and updates parameters accordingly. The post-training setting is attractive for large language models because full retraining is expensive, but the absence of retraining also makes the initial quantizer and any discrete refinement particularly important.

2.3 Related Work and Evidence Boundaries

General Post-training Weight Quantization

Post-training quantization methods for large language models differ in how they select ranges, account for activations, and compensate for local error. GPTQ [7] formulates layer-wise weight quantization using approximate second-order information and sequential error compensation. It is more precise to describe GPTQ as a block-wise reconstruction method than as a method that treats an entire layer with one homogeneous scale. Its objective accounts for calibration-informed curvature approximations within the processed layer, making it a strong general baseline for low-bit weight quantization.

AWQ [8] uses activation statistics to identify salient weight channels and applies scaling transformations intended to protect their contributions. AWQ therefore does not ignore differences among channels, and criticism of it must not imply that it uses only a global layer scale. Its salience criterion and scaling strategy address a different problem from explicitly evaluating coupled query–key score perturbations or the consequences of shared K/V heads in GQA. The existence of this difference motivates investigation of attention-specific refinement, but does not establish that AWQ is generally inadequate.

SmoothQuant [9] addresses weight-and-activation quantization by transferring quantization difficulty between activations and weights through mathematically equivalent scaling. It is especially relevant to the distinction between activation outliers and weight outliers: large activation features can be managed by changing the balance between the two operands. The present thesis leaves activations in higher precision, so SmoothQuant provides conceptual context rather than a directly matched weight-only baseline.

These methods demonstrate that reconstruction objectives, activation statistics, and local transformations can preserve model quality at low precision. They also illustrate why broad statements that existing methods are merely layer-wise or statistically unaware are inaccurate. The narrower research gap is that a general reconstruction objective need not explicitly optimize the interaction of Q and K projections, distinguish the reuse pattern induced by GQA, or revisit selected integer levels after initial group-wise asymmetric quantization. Whether those omissions are consequential is an empirical question rather than an assumption.

Adjacent Evidence from KV-cache Quantization

KVQuant [10] and QAQ [11] examine quantization associated with key and value cache representations. Their results are relevant because the KV cache participates directly in attention over previously processed tokens, and long contexts repeatedly reuse stored representations. However, KV-cache quantization and QKV projection-weight quantization intervene at different locations. Cache quantization approximates representations after projection and stores those approximations across decoding steps. Projection-weight quantization approximates the matrices that generate new queries, keys, and values from each input representation.

The two settings share downstream attention equations, so KV-cache evidence supports the general proposition that perturbations to keys and values can matter. It does not directly prove that projection weights are more sensitive than other weights, that a specific QKV projection must receive special treatment, or that a cache-oriented technique transfers to group-wise asymmetric weight-only quantization. Such conclusions require direct analysis or evaluation in the projection-weight setting. Treating KVQuant and QAQ as adjacent rather than direct evidence preserves their relevance without extending their claims beyond their experimental object.

Attention Architectures

Standard multi-head attention (MHA) gives each attention head its own query, key, and value head [12]. The separate heads can learn different relations and produce different attention distributions. This structure does not guarantee that perturbations remain isolated, because head outputs are later combined and transformed, but head-specific K/V representations do limit direct sharing at the score-computation stage.

Multi-query attention (MQA) shares one key head and one value head across many query heads [13]. GQA occupies an intermediate design point: several query heads are assigned to each shared key/value pair [14]. This reduces KV-cache size and can improve inference efficiency while retaining more K/V diversity than MQA. In GQA terminology, the query heads assigned to one shared key/value pair form a GQA group.

Sharing changes the dependency structure of perturbations. If a shared key is perturbed, every query head associated with that key computes scores using the same perturbed representation. Likewise, a shared value perturbation enters the weighted aggregation for every associated query head. The resulting errors are dependent because they contain a common component. Dependence does not imply that their magnitudes add monotonically or that the final model error is necessarily larger than in MHA. Query directions can make the shared key perturbation influential for some heads and nearly orthogonal for others; softmax can attenuate or redistribute score changes; value effects can cancel after output projection; and later layers can suppress or compound the perturbation. GQA therefore creates a pathway for conditional amplification and coordinated error, not a universal amplification law.

2.4 Group-wise Asymmetric Weight-only Quantization

Local Representation and Residual Error

The purpose of group-wise quantization is to adapt the affine range to local weight statistics. A projection matrix can contain rows and regions with different minima, maxima, variances, and tail behavior. If all regions share one scale, a few extreme values can enlarge the step size for many otherwise compact weights. Partitioning each output-channel row along input channels reduces the number of weights competing for one range. Asymmetric parameters further permit a local interval that is not centered at zero, while the levels remain uniformly spaced.

This granularity does not eliminate quantization error. Within a group, every weight still uses the same step size and endpoints. A single extreme weight can expand the range of its group. Alternatively, clipping or range selection can preserve finer resolution for the bulk of the group while producing larger endpoint errors for excluded values. The preferable trade-off depends on the objective. Minimizing unweighted element-wise error may favor one range, whereas minimizing calibration-output error may favor another because input features weight the columns unequally.

For a group g with scale s_g and zero-point z_g, adjacent-level adjustment changes a selected integer c_i to $c_i + 1$ or $c_i - 1$, provided the result remains within the representable interval. The corresponding dequantized weight changes by $+s_g$ or $-s_g$. The operation leaves all other integer values and the group's affine parameters unchanged. This local discreteness makes the effect easy to define, but not automatically beneficial. Moving to an adjacent level normally increases the selected weight's nearest-rounding error unless the initial assignment was altered by clipping or another objective. Its justification must therefore come from a broader reconstruction or interaction-aware criterion.

The phrase bit-level adjustment is used for this adjacent-level operation throughout the thesis. It does not mean toggling a literal binary bit, changing INT4 to another bit width, or introducing a special high-precision exception. A binary representation is only an encoding of the integer level; the semantic operation is movement to a neighboring integer code. This definition also makes adjustment independent of whether signed integers, offset integers, or another equivalent low-bit encoding is used.

Local and Downstream Objectives

Quantization quality can be measured at several levels. Weight reconstruction compares $\hat{W}$ with W, often using a norm such as $\|\Delta W\|_F^2$. Projection reconstruction evaluates $\|X\Delta W^\top\|_F^2$ on calibration inputs. Attention-aware objectives can compare score matrices, probability matrices, or attention outputs. These objectives are related but not equivalent.

An element-wise objective treats every weight error equally. A projection-output objective weights errors according to the calibration activations and their correlations. A score objective couples Q and K errors because the score matrix is bilinear. An attention-output objective additionally includes softmax sensitivity and value perturbations. Moving to a more downstream objective can better represent the target behavior, but it also becomes more input-dependent and may be more expensive or susceptible to calibration overfitting. A selective refinement method therefore needs an explicitly stated objective and cannot assume that reducing one error measure reduces every later error measure.

The initial group-wise asymmetric quantizer supplies a regular low-bit representation. Selective adjacent-level refinement can then revisit a restricted set of integer assignments under a chosen criterion. Selection is necessary because exhaustively evaluating both neighboring levels for every weight, especially in many large projection matrices, is expensive. Candidate selection may use rounding proximity, local error, distribution summaries, or estimated downstream influence. None of these indicators alone proves that an adjustment will improve model behavior; they narrow the search space for subsequent evaluation.

2.5 Attention Computation and Projection Perturbations

Notation and Projection Orientation

Let $X \in \mathbb{R}^{n \times d_{\text{in}}}$ contain n token representations as rows. Let $W_Q \in \mathbb{R}^{d_Q \times d_{\text{in}}}$, $W_K \in \mathbb{R}^{d_K \times d_{\text{in}}}$, and $W_V \in \mathbb{R}^{d_V \times d_{\text{in}}}$. For a single row vector x, the projections are

$$q = xW_Q^\top, \qquad k = xW_K^\top, \qquad v = xW_V^\top. \tag{2.3}$$

Applied to all tokens, the corresponding matrices are

$$Q = XW_Q^\top, \qquad K = XW_K^\top, \qquad V = XW_V^\top. \tag{2.4}$$

This orientation is used consistently: rows of each weight matrix are output channels, columns are input channels, and group-wise quantization partitions columns within each row.

For one attention head, scaled dot-product attention is

$$S = \frac{QK^\top}{\sqrt{d_k}}, \qquad P = \mathrm{softmax}_{\text{row}}(S), \qquad O = PV, \tag{2.5}$$

where d_k is the head dimension, S is the score matrix, P contains row-wise attention probabilities, and O is the weighted aggregation of values [12]. Multi-head variants apply this process across heads and subsequently combine their outputs.

From Weight Error to Projection Error

Let the quantized projection weights be

$$\hat{W}_Q = W_Q + \Delta W_Q, \qquad \hat{W}_K = W_K + \Delta W_K, \qquad \hat{W}_V = W_V + \Delta W_V. \tag{2.6}$$

The induced projection perturbations are

$$\Delta Q = X\Delta W_Q^\top, \qquad \Delta K = X\Delta W_K^\top, \qquad \Delta V = X\Delta W_V^\top, \tag{2.7}$$

so that $\hat{Q} = Q + \Delta Q$, $\hat{K} = K + \Delta K$, and $\hat{V} = V + \Delta V$. The transpose is essential because the rows of each weight matrix are output channels. These equations also make the role of activations explicit: the same ΔW_Q, ΔW_K, or ΔW_V produces different projection errors for different X.

The perturbed unscaled score product expands as

$$\hat{Q}\hat{K}^\top = QK^\top + \Delta QK^\top + Q\Delta K^\top + \Delta Q\Delta K^\top. \tag{2.8}$$

After division by $\sqrt{d_k}$, the score perturbation is therefore

$$\Delta S = \frac{\Delta QK^\top + Q\Delta K^\top + \Delta Q\Delta K^\top}{\sqrt{d_k}}. \tag{2.9}$$

The first two terms are first order in the projection perturbations, while the last term is second order. When perturbations are small, the second-order term may be less influential, but low bit widths, clipping, large activation directions, or aligned errors can make it non-negligible. The expansion shows why independently small Q and K reconstruction errors do not determine score error by themselves: orientation and interaction with the unperturbed projections also matter.

A norm bound illustrates possibilities without predicting realized behavior:

$$\|\Delta S\| \leq \frac{\|\Delta Q\|\,\|K\| + \|Q\|\,\|\Delta K\| + \|\Delta Q\|\,\|\Delta K\|}{\sqrt{d_k}}, \tag{2.10}$$

for compatible submultiplicative norms. This upper bound can be loose because it discards direction and cancellation. It establishes that large projection errors can permit large score error, but it does not prove that the bound is attained or that QKV weights are universally more sensitive than other model parameters.

Local Sensitivity of Softmax

Softmax converts each score row into a probability distribution. For a score vector s and $p = \mathrm{softmax}(s)$, the Jacobian is

$$J_{\mathrm{softmax}}(s) = \mathrm{diag}(p) - pp^\top. \tag{2.11}$$

For a sufficiently small score perturbation δs, the local first-order probability change is approximately

$$\delta p \approx \left(\mathrm{diag}(p) - pp^\top\right) \delta s. \tag{2.12}$$

This expression provides a more precise account than the blanket statement that softmax amplifies errors. Softmax is invariant to adding the same constant to all elements of one score row, so that perturbation direction has no effect. When one probability is already close to one, some perturbations may be strongly attenuated. When several scores are close and receive substantial probability, perturbations that alter their relative differences can redistribute attention meaningfully. The effect is therefore local, directional, and dependent on the current score distribution.

For finite perturbations, a change can cross a region in which the local Jacobian itself changes. A small score error near a narrow decision margin may alter which token receives the largest probability, but even a ranking change need not cause a large output change if the corresponding value vectors are similar. Conversely, a modest probability redistribution can matter if it transfers weight between very different values. Softmax is thus one conditional link in the propagation chain rather than a universal amplifier.

Value Error and Attention-output Error

With $\hat{P} = P + \Delta P$ and $\hat{V} = V + \Delta V$, the attention-output perturbation is

$$\hat{O} - O = \Delta P V + P \Delta V + \Delta P \Delta V. \tag{2.13}$$

The term $\Delta P V$ reflects changes in attention probabilities, $P \Delta V$ reflects direct value-projection error under the original probabilities, and $\Delta P \Delta V$ couples both effects. This decomposition distinguishes the roles of Q/K and V. Q and K affect scores and probabilities; V affects the content being aggregated. A large score perturbation can have limited output effect when relevant values are similar, while a small value perturbation can matter when it is repeatedly assigned high attention probability.

Later operations, including head concatenation, output projection, residual addition, normalization, and subsequent layers, can attenuate, preserve, or compound the attention-output perturbation. It is therefore defensible to say that QKV quantization creates structured pathways for downstream error, but not that every local QKV error necessarily produces a proportionally larger final error. Claims of comparative sensitivity require direct measurements under specified models, data, quantizers, and evaluation tasks.

2.6 GQA and Dependent Error

GQA changes how query heads are mapped to key and value heads. Suppose query heads $h \in \mathcal{G}_r$ belong to GQA group r and share K_r and V_r. For each associated query head,

$$S_h = \frac{Q_h K_r^\top}{\sqrt{d_k}}, \qquad O_h = \mathrm{softmax}_{\mathrm{row}}(S_h)V_r. \tag{2.14}$$

If the shared key is perturbed by ΔK_r, then every $h \in \mathcal{G}_r$ contains the score-error term

$$\frac{Q_h \Delta K_r^\top}{\sqrt{d_k}}. \tag{2.15}$$

These errors are dependent because the same ΔK_r appears in every term. Their realized magnitudes and signs differ because the Q_h differ. Similarly, a shared ΔV_r enters the output of every associated query head, but each head weights that value perturbation with its own attention probabilities.

Dependence matters for analysis because an independence assumption would overlook common error sources. It does not imply coherent accumulation. If query directions differ, some products with ΔK_r may be small or oppose one another after later transformations. If attention distributions differ, the same ΔV_r may be weighted differently. The output projection can mix heads in a manner that cancels or compounds their errors. Consequently, GQA can increase the reach of a shared perturbation and can conditionally amplify its downstream effect, yet the architecture alone does not determine the final magnitude.

Comparisons among MHA, GQA, and MQA should therefore be stated structurally. MHA uses more independent K/V heads; MQA shares K/V most broadly; GQA shares within intermediate-sized architectural groups. Broader sharing creates more query heads that depend on each shared K/V perturbation. Whether this produces worse model quality after quantization depends on the learned weights, input activations, quantization errors, softmax operating points, and downstream mixing. The thesis focuses on GQA because shared K/V makes interaction-aware analysis especially relevant, not because GQA is assumed to be universally fragile.

The numerical quantization groups used in the thesis remain independent of this architecture. A quantization group lies inside one output-channel row and spans contiguous input channels. A GQA group associates several query heads with a shared key/value head. The two kinds of groups can interact indirectly because quantization error in weights contributing to a shared head is reused architecturally, but neither group definition determines the other.

2.7 Outliers, Salience, and Candidate Selection

Weight Outliers and Quantization Range

A weight outlier is a weight whose magnitude or value is extreme relative to an explicitly specified reference population, such as its quantization group. The reference population matters: a value can be ordinary at the layer level but extreme within a small local group, or the reverse. Under min–max affine quantization, extreme endpoints influence the scale. Preserving a distant endpoint can enlarge the step size and increase rounding error for many central values. Clipping the endpoint can reduce the step size but incurs a potentially large clipping error for the excluded weight [5], [15].

At INT4, the limited number of levels makes this trade-off visible. Nevertheless, the presence of a weight outlier does not establish that it should be clipped, preserved exactly, or selected for adjustment. Its functional effect depends on the input feature multiplying it, correlations with other errors, and the downstream objective. Group-wise affine quantization localizes the range trade-off, but a single extreme weight can still affect every weight sharing its numerical group.

Activation Outliers and Functional Salience

An activation outlier is an unusually large or otherwise extreme activation feature under a specified input distribution. It is conceptually distinct from a weight outlier. Weight-only quantization does not directly approximate activations, but activation magnitude influences how weight errors are expressed because projection error equals $X\Delta W^{\top}$. A modest weight error in a frequently large activation direction can have a substantial projection effect. SmoothQuant [9] and AWQ [8] illustrate, in different settings, why activation statistics are relevant to quantization decisions.

Salience refers to functional importance under a stated criterion. A weight can be salient because changing it strongly affects calibration outputs or another objective, even if its magnitude is ordinary. Conversely, a large weight or activation can have limited effect under a particular data distribution or can be compensated elsewhere. Outlier status is distributional; salience is objective-dependent. Treating the terms as synonyms would obscure the difference between selecting extreme values and selecting influential values.

Candidate-selection heuristics can combine these signals, but their roles should remain explicit. Distance to a quantization boundary indicates rounding ambiguity.

Magnitude indicates extremeness relative to a reference set. Calibration-weighted error estimates realized projection influence on the calibration sample. Interaction-aware measures estimate effects on score or attention-output objectives. Each signal can narrow the candidate set, but final benefit must be assessed using the selected refinement objective and, ultimately, model evaluation.

Kneedle as a Heuristic

Kneedle was introduced as a general method for locating knee points in suitable curves [1]. A knee is a region where the curve's rate of improvement or growth changes markedly after normalization and comparison with a reference line. The algorithm's general literature role is curve-based transition-point identification. It is not inherently an outlier detector, a quantization algorithm, or a proof that a selected threshold is optimal.

In a quantization context, one may construct a monotone summary from sorted weight magnitudes, cumulative energy, reconstruction error, or another quantity and apply Kneedle to propose a transition point. The result depends on the constructed curve, normalization, smoothing, and sensitivity parameters. Applying the algorithm to a weight summary can therefore yield a reproducible heuristic threshold or candidate budget, but the values beyond that point are not thereby proven to be harmful outliers or salient weights.

This distinction separates the general literature claim from the later methodology of the thesis. The literature supports Kneedle as a knee-point heuristic. Any use of its output to choose candidates for adjacent-level adjustment is a methodological design choice that must be evaluated under the quantization objective. The method may help avoid a fixed global threshold when groups have different distributions, but it does not guarantee a better scale, a better candidate set, or improved downstream accuracy.

2.8 Selective Adjacent-level Refinement

After initial group-wise asymmetric quantization, each weight is represented by an integer level and a group-specific affine transformation. A selective refinement stage can evaluate whether moving a candidate integer to an adjacent level improves a chosen objective. If c_i is the stored integer for weight i in group g, the candidate set is restricted to valid neighbors,

$$\mathcal{N}(c_i) = \{c_i - 1, c_i + 1\} \cap [c_{\min}, c_{\max}]. \tag{2.16}$$

Changing c_i to an element of $\mathcal{N}(c_i)$ is the bit-level adjustment considered by this thesis. It moves the dequantized value by exactly one local step s_g. It is not a literal binary-digit flip and does not alter s_g, z_g, group membership, or bit width.

Restricting adjustment to adjacent levels has several conceptual advantages. It bounds the local change, preserves the original quantizer, and permits a clear comparison between the current assignment and its immediate alternatives. It also avoids interpreting a large jump caused by flipping a high-order binary digit as a small refinement. However, adjacency is a constraint, not evidence of improvement. A neighbor can reduce an interaction-aware objective even while increasing element-wise weight error, or it can worsen both. Every accepted adjustment therefore requires an explicit acceptance criterion.

Sequential adjustment also introduces interactions among decisions. The benefit estimated for one candidate can change after another candidate is moved because projection, score, and attention-output errors contain sums and cross terms. Evaluating candidates independently is efficient but may miss coordinated effects. Evaluating all combinations is generally impractical. A selective procedure occupies the middle ground: use heuristics to restrict candidates, evaluate adjacent alternatives according to a stated objective, and recognize that the resulting solution is a local discrete refinement rather than a proven global optimum.

The choice of objective determines what the refinement preserves. A weight-space objective is inexpensive and input-independent but ignores activation directions. A projection-space objective uses calibration inputs and directly measures $X\Delta W^\top$. A score-space objective couples Q and K but omits the conditional transformation through softmax and values. An attention-output objective includes more of the relevant computation but is more input-dependent. No single objective guarantees improvement on all downstream tasks. The rationale for interaction-aware refinement is therefore that it measures a quantity closer to attention behavior, not that it universally dominates simpler reconstruction.

2.9 Statistical Interpretation and Its Limits

James–Stein Estimation

The James–Stein result concerns estimation of a multivariate normal mean under specific assumptions [2]. In its classical form, one observes

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2 I_d), \tag{2.17}$$

with known common variance, squared-error risk, and dimension $d \geq 3$. The James–Stein estimator shrinks the observation vector toward a chosen center, commonly the origin:

$$\hat{\boldsymbol{\theta}}_{\mathrm{JS}} = \left(1 - \frac{(d-2)\sigma^2}{\|\mathbf{x}\|^2}\right)\mathbf{x}. \tag{2.18}$$

Under the classical conditions and the relevant risk definition, shrinkage can dominate the component-wise maximum-likelihood estimator in total expected squared error. The result demonstrates that accepting coordinated bias can reduce aggregate estimation risk in a particular statistical model.

Quantized neural-network weights do not generally satisfy these assumptions. Their errors are discrete, bounded or clipped by a quantizer, grouped under shared affine parameters, multiplied by data-dependent activations, and evaluated through nonlinear model behavior. The unknown-mean Gaussian model, known isotropic variance, and classical squared-error risk do not directly describe this setting. James–Stein theory therefore does not prove that adjacent-level adjustment, Flip, or ReFlip improves a quantized model. It also does not prescribe which weights to adjust or which attention-aware objective to use.

The valid role of James–Stein reasoning in this literature review is modest and analogical. It cautions against assuming that independently optimal component decisions must minimize an aggregate criterion, and it illustrates how controlled bias can improve total risk under suitable assumptions. This analogy can motivate empirical investigation of coordinated or objective-aware decisions, but it is neither a principled basis in the sense of a derived guarantee nor evidence for a specific refinement procedure.

Connecting Heuristics to Evaluation

Kneedle and James–Stein reasoning occupy different roles. Kneedle is an algorithmic heuristic for proposing a transition point on a constructed curve. James–Stein is a statistical theorem under a specific probabilistic model that offers a limited analogy about aggregate risk. Neither establishes that a selected quantization candidate is salient, that an adjacent-level move is beneficial, or that a downstream model metric will improve.

A defensible refinement framework therefore separates proposal, acceptance, and validation. A heuristic may propose candidate weights or a candidate budget. A specified reconstruction or interaction-aware objective determines whether an adjacent-level alternative is accepted. Model-level evaluation determines whether the resulting quantized model preserves behavior on relevant data. This separation prevents a selection heuristic from being interpreted as proof and prevents an analogy from being interpreted as an optimization guarantee.

2.10 Synthesis and Research Gap

The literature establishes several relevant facts while leaving a focused gap. First, post-training quantization can compress large language models effectively without full retraining, and methods such as GPTQ and AWQ use substantially more information than a single layer-wide scale [7], [8]. Second, affine asymmetry and group-wise granularity address different aspects of representation: asymmetry shifts a uniformly spaced local range, while grouping localizes the parameters shared by weights. Third, attention couples projection errors through a bilinear score product, a locally sensitive softmax transformation, and value aggregation. Fourth, GQA causes several query heads to depend on shared K/V perturbations, creating dependent errors whose eventual amplification, attenuation, or cancellation is conditional.

The evidence does not support several stronger claims. KV-cache studies do not directly prove projection-weight sensitivity. Weight magnitude does not establish salience. Softmax does not universally amplify score perturbations. GQA sharing does not guarantee coherent accumulation. Kneedle does not prove that a threshold separates harmful outliers, and James–Stein theory does not prove the value of a discrete quantization refinement. Preserving these boundaries narrows the thesis claim but makes it testable.

Within these boundaries, the research gap is the selective refinement of group-wise asymmetric low-bit QKV projection weights under objectives that can account for more than independent element-wise rounding. The initial quantizer provides a regular, efficient representation. Bit-level adjustment, precisely understood as an adjacent integer-code or quantization-level change, provides a restricted discrete action after quantization. Distributional and calibration-based heuristics can reduce the candidate set, while projection- or interaction-aware objectives can assess candidate changes. The resulting method must be judged empirically; its motivation follows from the structure of the problem, not from a claim of universal sensitivity or a statistical guarantee.

Conclusion

This chapter has defined the thesis setting as post-training, group-wise asymmetric INT4 weight-only quantization of QKV projection weights. It distinguished input channels, output channels, numerical quantization groups, and architectural GQA groups; clarified that asymmetric affine quantization remains uniformly spaced; and defined bit-level adjustment as movement to an adjacent integer code and quantization level rather than a literal binary-digit flip.

The perturbation analysis showed that projection-weight errors induce $\Delta Q = X\Delta W_Q^\top$, $\Delta K = X\Delta W_K^\top$, and $\Delta V = X\Delta W_V^\top$. Query and key errors interact in the score product, after which softmax transforms score perturbations according to its local, directional sensitivity. GQA reuses shared K/V perturbations across associated query heads and therefore creates dependent errors, but their downstream accumulation or amplification remains conditional on queries, score margins, values, output mixing, and later computations.

Finally, the chapter established careful evidence boundaries. KVQuant and QAQ provide adjacent evidence about the KV cache rather than direct proof about projection weights. GPTQ and AWQ are strong, structured post-training methods, but their objectives differ from explicit refinement of attention interactions. Kneedle can guide candidate selection but does not certify outliers or optimal thresholds. James–Stein estimation offers only a restricted analogy under assumptions that do not directly hold for quantized neural networks. These conclusions motivate the later methodology while leaving its effectiveness to direct empirical evaluation.

3.1 Overview

This chapter presents a post-training, weight-only quantization methodology for the query, key, and value projections of transformer attention. The methodology is designed for asymmetric group-wise INT4 quantization: projection weights are stored on a four-bit integer grid with group-specific scales and zero points, whereas activations remain in their original precision. The central premise is that nearest rounding is a strong starting point but not necessarily the best final choice when the quantized weights are evaluated through the activations and interactions that determine attention.

The proposed refinement has two complementary stages. The first stage, called **Flip**, treats a projection matrix independently and reduces its activation-weighted output residual. Flip is deliberately distinguished from ordinary round-to-nearest (RTN) quantization. RTN already minimizes the isolated scalar reconstruction error of each weight on a fixed quantization grid. Consequently, changing a rounded integer to a neighboring level generally cannot improve that isolated scalar objective. Flip instead accepts a neighboring level only when the resulting change reduces the aggregate projected residual. Such a change may increase the error of an individual weight while improving the output of the complete projection through cancellation among activation-weighted errors.

The second stage, called **ReFlip**, addresses the coupling between query and key projections. ReFlip holds the quantized key matrix fixed and updates only the quantized query matrix. It evaluates how a legal one-step integer change affects a scalar query–key score formed from a representative activation. The resulting objective is exact for this scalar score: no Taylor approximation or full-matrix gradient surrogate is used. ReFlip can therefore compensate for a score residual jointly caused by query and key quantization, even though only the query weights are modified. This compensation is useful, but its scope must be stated precisely: preservation of one representative scalar score does not imply preservation of the full pairwise attention-score matrix over all token pairs.

Both refinement stages use a representative activation row vector obtained from calibration data and stabilized by James–Stein shrinkage. This produces a compact, data-dependent signal for evaluating discrete changes without retraining. Kneedle-based selection is used in ReFlip to identify an extreme high-sensitivity region that should be protected and a moderate-sensitivity region immediately after the knee in which correction is permitted. Kneedle is an adaptive heuristic rather than an optimality guarantee.

The overall procedure remains fully post-training. It requires neither gradient-based optimization nor modification of the model architecture. The refinement is performed before deployment, and the resulting matrices retain the same group-wise asymmetric INT4 representation used by the initial quantizer. Thus, the method aims to improve the functional fidelity of quantized attention projections while preserving the principal memory motivation of low-bit weight storage and without making claims about measured runtime.

3.2 Notation and Calibration Representative

Projection Convention

Let a representative activation be a row vector

$$x \in \mathbb{R}^{d_{\text{in}}}. \tag{3.1}$$

For each attention projection, the full-precision weight matrix has shape

$$W_Q, W_K, W_V \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}. \tag{3.2}$$

The corresponding projected row vectors are

$$q = xW_Q^\top, \qquad k = xW_K^\top, \qquad v = xW_V^\top, \tag{3.3}$$

where $q, k, v \in \mathbb{R}^{d_{\text{out}}}$. This row-vector convention is used throughout the chapter. In particular, matrix row a identifies an output dimension and matrix column b identifies an input dimension.

The quantized and dequantized projection matrices are denoted by

$$\hat{W}_Q, \hat{W}_K, \hat{W}_V \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}, \tag{3.4}$$

with projected vectors

$$\hat{q} = x\hat{W}_Q^\top, \qquad \hat{k} = x\hat{W}_K^\top, \qquad \hat{v} = x\hat{W}_V^\top. \tag{3.5}$$

Although the stored representation is integer-valued, the hat notation refers to the dequantized value represented by those integers, scales, and zero points.

James–Stein-Shrunk Activation

Calibration activations can be noisy when only a limited post-training sample is available. A single unregularized statistic may overemphasize accidental extremes, whereas a simple global average may suppress meaningful differences between input dimensions. To obtain a stable representative row vector, this work uses a positive-part James–Stein shrinkage estimate.

Let $\tilde{x} \in \mathbb{R}^{d_{\text{in}}}$ be an empirical activation statistic computed from the calibration set, and let $m \in \mathbb{R}^{d_{\text{in}}}$ be a shrinkage target. A common target is a constant vector whose entries equal the grand mean of $\tilde{x}$, although another predetermined calibration target may be used. The representative activation is

$$x = m + \lambda(\tilde{x} - m), \tag{3.6}$$

where the positive-part shrinkage factor has the form

$$\lambda = \left[1 - \frac{(d_{\text{in}} - 2)\hat{\sigma}^2}{\|\tilde{x} - m\|_2^2}\right]_+, \qquad [u]_+ = \max(u, 0). \tag{3.7}$$

Here, $\hat{\sigma}^2$ estimates calibration noise. If the observed dimension-wise variation is weak relative to the estimated noise, λ moves the statistic toward m; if the observed variation is strong, more of the empirical structure is retained. The resulting x is not claimed to represent every token or every sequence. It is a compact calibration summary used to define the exact objectives evaluated by Flip and ReFlip.

This distinction is important. The refinement stages optimize behavior under the chosen representative activation. Their objectives can improve a targeted calibration criterion, but they do not provide a universal guarantee over the complete activation distribution.

3.3 Problem Formulation

The goal is to construct group-wise asymmetric INT4 approximations of the attention projection matrices while limiting functionally relevant quantization error. Conventional RTN quantization primarily considers each scalar weight on its own. Attention, however, is determined by projected activations and by interactions between the resulting query and key vectors.

For a generic projection matrix W, define the dequantization error

$$E = \hat{W} - W. \tag{3.8}$$

Under the representative activation, the corresponding projection residual is

$$r = xE^\top = x(\hat{W} - W)^\top \in \mathbb{R}^{d_{\text{out}}}. \tag{3.9}$$

The residual in output dimension j is therefore

$$r_j = \sum_{i=1}^{d_{\text{in}}} x_i E_{ji}. \tag{3.10}$$

This expression makes the relevant error aggregation explicit. Individual signed weight errors are multiplied by activation coordinates and then summed. Two nonzero scalar weight errors can partially cancel in the projected output, while a small scalar error can matter substantially when multiplied by a large activation coordinate.

The local functional objective used by Flip is

$$J(W, \hat{W}; x) = \|r\|_2^2 = \left\|x(\hat{W} - W)^\top\right\|_2^2. \tag{3.11}$$

This objective is applicable independently to W_Q, W_K, and W_V. It measures projection fidelity for the representative activation and does not claim to optimize the full attention computation.

The interaction-aware objective used by ReFlip is narrower and more explicit. Define the original scalar query–key score

$$s^\star = (xW_Q^\top) \cdot (xW_K^\top) = q \cdot k, \tag{3.12}$$

and the current quantized score

$$\hat{s} = (x\hat{W}_Q^\top) \cdot (x\hat{W}_K^\top) = \hat{q} \cdot \hat{k}. \tag{3.13}$$

ReFlip minimizes

$$\mathcal{L} = \frac{1}{2}(\hat{s} - s^\star)^2. \tag{3.14}$$

The scalar residual

$$e_s = \hat{s} - s^\star \tag{3.15}$$

contains the combined effect of quantizing both query and key projections. ReFlip holds $\hat{W}_K$ fixed and changes only $\hat{W}_Q$, using the latter as the available degree of freedom for reducing this joint score residual.

These two objectives establish a progression from projection-level fidelity to query–key interaction fidelity. Flip minimizes an activation-weighted residual for one matrix at a time. ReFlip then compensates the remaining scalar query–key score residual through legal integer changes to $\hat{W}_Q$. Neither objective is presented as an exact surrogate for all downstream model behavior.

3.4 Group-Wise Asymmetric INT4 Quantization

Integer Representation

For a generic projection matrix

$$W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}, \tag{3.16}$$

the input dimension is partitioned into groups. Let $g(b)$ map input column b to its group. Each output row and input group has a scale

$$\alpha_{a,g(b)} > 0 \tag{3.17}$$

and a zero point

$$\zeta_{a,g(b)}. \tag{3.18}$$

The stored INT4 value at row a and column b is

$$z_{ab} \in \mathcal{Z}_4 = \{0, 1, \ldots, 15\}. \tag{3.19}$$

Its dequantized value is

$$\hat{W}_{ab} = \alpha_{a,g(b)} \left(z_{ab} - \zeta_{a,g(b)} \right). \tag{3.20}$$

For each row-group pair, the initial asymmetric scale and zero point are derived from the local weight range. With $w_{\min}$ and $w_{\max}$ denoting the extrema in that row-group,

$$\alpha = \frac{w_{\max} - w_{\min}}{2^4 - 1}, \tag{3.21}$$

followed by an appropriately clipped zero point. The initial RTN integer is

$$z_{ab}^{\text{RTN}} = \text{clip}_{[0,15]} \left(\text{round} \left(\frac{W_{ab}}{\alpha_{a,g(b)}} + \zeta_{a,g(b)} \right) \right). \tag{3.22}$$

Degenerate groups with zero range require a predetermined positive scale convention, but no refinement is useful unless at least two dequantized levels are distinguishable.

What RTN Optimizes

For fixed $\alpha_{a,g(b)}$ and $\zeta_{a,g(b)}$, RTN chooses the legal quantization level closest to W_{ab}. Except for ties and clipping boundaries, it solves the isolated scalar problem

$$\min_{z\in\mathcal{Z}_4} \left|\alpha_{a,g(b)}\left(z-\zeta_{a,g(b)}\right)-W_{ab}\right|. \tag{3.23}$$

Equivalently, it minimizes the squared scalar reconstruction error on the fixed grid. Therefore, a neighboring integer change should not be described as improving the isolated scalar weight error that RTN already minimizes.

The limitation of RTN is instead an objective mismatch. The projected residual in output row a depends on the signed sum

$$r_a = \sum_b x_b(\hat{W}_{ab} - W_{ab}), \tag{3.24}$$

not on each absolute scalar error separately. A neighboring integer can increase

$$|\hat{W}_{ab} - W_{ab}| \tag{3.25}$$

yet reduce $|r_a|$ because its signed activation-weighted contribution better cancels the contributions of other weights. Flip exploits precisely this possibility.

Memory and Inference Motivation

INT4 storage provides the theoretical basis for reducing the weight memory of QKV projections. Ignoring metadata, four-bit weights require one quarter of the storage of FP16 weights and one eighth of the storage of FP32 weights. Group-wise scales and zero points introduce overhead, so the exact compression ratio depends on group size and metadata precision. Smaller groups can improve local representational fidelity but increase this overhead.

The refinement stages do not change the deployed tensor shapes or the mathematical dequantization rule. They modify legal stored integers while retaining the same scales and zero points. Consequently, the refined model remains compatible with the same class of group-wise weight-only inference procedure as its RTN initialization. Because the present methodology does not report a measured runtime study, it makes no empirical latency or throughput claim. Its inference motivation is structural: low-bit weight storage can reduce memory traffic and capacity requirements, while the offline refinement seeks to recover fidelity without introducing a new runtime objective or a new projection operator.

3.5 Flip: Activation-Weighted Projection Refinement

Flip Objective

For one projection matrix W and its current quantized approximation $\hat{W}$, Flip minimizes

$$J = \|r\|_2^2, \qquad r = xE^\top, \qquad E = \hat{W} - W. \tag{3.26}$$

The objective separates across output rows:

$$J = \sum_{j=1}^{d_{\text{out}}} r_j^2. \tag{3.27}$$

Nevertheless, it does not separate across the weights within a row because all input coordinates contribute to the same scalar residual r_j. This within-row coupling is the source of useful error cancellation.

Suppose a legal candidate changes only the dequantized weight at row j and column i by

$$\delta = \hat{W}'_{ji} - \hat{W}_{ji}. \tag{3.28}$$

For a one-step integer move,

$$\delta = \alpha_{j,g(i)}\Delta z, \qquad \Delta z \in \{-1, +1\}, \tag{3.29}$$

provided that $z_{ji} + \Delta z \in \mathcal{Z}_4$. The candidate changes only residual component r_j:

$$r'_j = r_j + x_i\delta. \tag{3.30}$$

All other residual components remain unchanged. The exact objective change is consequently

$$\begin{aligned} \Delta J &= J' - J && (3.31)\\ &= (r_j + x_i\delta)^2 - r_j^2 && (3.32)\\ &= 2r_jx_i\delta + (x_i\delta)^2. && (3.33) \end{aligned}$$

No approximation is required. A legal candidate is accepted only if

$$\Delta J < 0. \tag{3.34}$$

After acceptance, the scalar residual for that output row is updated as

$$r_j \leftarrow r_j + x_i \delta. \tag{3.35}$$

This incremental update preserves exact agreement with the current quantized matrix while avoiding repeated reconstruction of the full residual.

Interpretation as Error Cancellation

The sign of the first term in ΔJ determines whether a candidate initially moves against the current residual:

$$2 r_j x_i \delta < 0. \tag{3.36}$$

The second term,

$$(x_i \delta)^2, \tag{3.37}$$

penalizes an excessively large move and prevents a candidate from being accepted merely because it has the correct direction. The exact acceptance condition can also be written as

$$|r_j + x_i \delta| < |r_j|. \tag{3.38}$$

Thus, Flip selects a neighboring quantization level only when its activation-weighted contribution reduces the magnitude of the aggregate row residual.

This interpretation resolves an apparent contradiction between RTN and Flip. RTN is already optimal for the isolated scalar reconstruction error on the fixed grid. Flip does not overturn that fact. Instead, it changes the objective from isolated scalar error to activation-weighted projection error. Consider a row containing two signed weight errors whose activation-weighted contributions reinforce one another. Moving one weight away from its nearest level can reverse or weaken one contribution, producing greater cancellation in their sum. The modified weight can be less accurate in isolation while the projected output becomes more accurate.

Accordingly, Flip must not be characterized as a second nearest-rounding search. It is a discrete coordinate refinement under the exact aggregate objective J. Its benefit depends on the representative activation and on the signed configuration of errors within each output row.

Candidate Evaluation and Acceptance

Flip begins from the RTN-quantized matrix and its exact residual r. For each eligible integer z_{ji}, it evaluates the legal directions in

$$\mathcal{D}_{ji} = \left\{ \Delta z \in \{-1, +1\} : z_{ji} + \Delta z \in \mathcal{Z}_4 \right\}. \tag{3.39}$$

For each direction, it obtains $\delta = \alpha_{j,g(i)}\Delta z$ and evaluates the exact ΔJ. If neither direction decreases J, the integer remains unchanged. If both directions decrease J, the direction with the more negative ΔJ is preferred. Once a change is accepted, both z_{ji} and r_j are updated before subsequent candidates are evaluated.

Updating the residual after every accepted change is essential. Candidate gains are not independent: a previous change alters the cancellation needed in row j, and therefore alters the exact value of every later ΔJ in that row. Evaluating all candidates against a stale initial residual would not implement the stated greedy objective.

Candidate order is a heuristic choice. One option is to evaluate coordinates in descending order of the magnitude of their possible activation-weighted step,

$$|x_i|\alpha_{j,g(i)}. \tag{3.40}$$

Another is to scan deterministically by row and column. The acceptance equation remains exact under either ordering, but the final discrete configuration can differ because accepted changes modify later residuals. A pass budget and an acceptance budget are therefore explicit hyperparameters rather than hidden assumptions.

Scope and Limitations of Flip

Flip can be applied independently to the query, key, and value projection matrices because its objective requires only a matrix, its quantized approximation, and the representative activation. For W_Q and W_K, reducing the projection residual can indirectly improve the vectors used in query–key scoring. For W_V, it can improve the representative value projection. However, Flip does not directly optimize the interaction between query and key, the softmax operation, or the final attention output.

The objective is also calibration-dependent. A change that reduces J under the representative James–Stein-shrunk vector may not reduce projection error for every activation row. The shrinkage estimate and conservative integer neighborhood are intended to limit sensitivity to calibration noise, but they do not constitute a distribution-wide guarantee. Finally, the greedy coordinate procedure can stop at a discrete local configuration even when a coordinated multi-weight change would yield a lower objective.

3.6 ReFlip: Scalar Query–Key Score Refinement

Fixed-Key Objective

After initial quantization and optional Flip refinement, ReFlip targets the scalar query–key score residual. The original score is

$$s^\star = (xW_Q^\top) \cdot (xW_K^\top), \tag{3.41}$$

and the current quantized score is

$$\hat{s} = (x\hat{W}_Q^\top) \cdot (x\hat{W}_K^\top). \tag{3.42}$$

Define

$$e_s = \hat{s} - s^\star, \qquad \mathcal{L} = \frac{1}{2}e_s^2. \tag{3.43}$$

During ReFlip, $\hat{W}_K$ is held fixed. ReFlip updates $\hat{W}_Q$ only; it does not update $\hat{W}_K$ or $\hat{W}_V$. This asymmetric choice makes the objective and update rule unambiguous. Although the available changes belong only to the query matrix, the target residual e_s includes quantization effects from both W_Q and W_K. ReFlip may therefore use a query-weight change to compensate the global Q/K score residual produced by the pair of quantized projections.

Here, "global" refers to the combined scalar query–key score rather than an isolated query-weight reconstruction error. It does not mean that the complete attention-score matrix is optimized. For a sequence with multiple query and key rows, pairwise scores depend on many activation pairs. Optimizing the scalar score associated with the representative x does not preserve, and cannot guarantee preservation of, the full pairwise attention matrix.

Exact Effect of an Integer Change

Consider query weight (a, b). ReFlip directly changes its stored integer:

$$z_{ab} \leftarrow z_{ab} + \Delta z_{ab}, \qquad \Delta z_{ab} \in \{-1, +1\}, \tag{3.44}$$

subject to the legal INT4 range

$$z_{ab} + \Delta z_{ab} \in \mathcal{Z}_4. \tag{3.45}$$

Because the scale and zero point are fixed, the dequantized query-weight change is

$$\Delta\hat{W}_{Q,ab} = \alpha_{a,g(b)}\Delta z_{ab}. \tag{3.46}$$

Only query output coordinate $\hat{q}_a$ changes, by

$$\Delta\hat{q}_a = x_b\alpha_{a,g(b)}\Delta z_{ab}. \tag{3.47}$$

Since $\hat{k}$ is held fixed, the exact scalar-score change is

$$\Delta s_{ab} = \hat{k}_a x_b \alpha_{a,g(b)}\Delta z_{ab}. \tag{3.48}$$

This equation is exact for the stated scalar objective. It is not a first-order Taylor approximation and does not require a gradient of a full attention matrix. The exact loss change is

$$\begin{aligned}
\Delta\mathcal{L}_{ab} &= \frac{1}{2}(e_s + \Delta s_{ab})^2 - \frac{1}{2}e_s^2 && (3.49)\\
&= e_s\Delta s_{ab} + \frac{1}{2}(\Delta s_{ab})^2 && (3.50)\\
&= (\hat{s} - s^\star)\Delta s_{ab} + \frac{1}{2}(\Delta s_{ab})^2. && (3.51)
\end{aligned}$$

A candidate is accepted only if

$$\Delta\mathcal{L}_{ab} < 0. \tag{3.52}$$

After an accepted integer change, the scalar score and residual are updated:

$$\hat{s} \leftarrow \hat{s} + \Delta s_{ab}, \qquad e_s \leftarrow e_s + \Delta s_{ab}. \tag{3.53}$$

The residual update must occur immediately because every subsequent exact loss change depends on the current e_s.

Parameter Sensitivity

The magnitude of the score change per unit integer step defines the parameter sensitivity:

$$\mathcal{S}_{ab} = \left|\frac{\Delta s_{ab}}{\Delta z_{ab}}\right| = \left|\hat{k}_a x_b \alpha_{a,g(b)}\right|. \tag{3.54}$$

This quantity combines three factors. The term $|\hat{k}_a|$ measures how strongly query output dimension a participates in the current scalar dot product. The term $|x_b|$ measures the contribution of input coordinate b under the representative activation. The scale $\alpha_{a,g(b)}$ gives the dequantized distance produced by a one-integer move.

Parameter sensitivities are aggregated over input columns by an ℓ_2 norm to define output-dimension sensitivity:

$$\mathcal{S}_a = \left(\sum_b \mathcal{S}_{ab}^2 \right)^{1/2} \tag{3.55}$$

$$= |\hat{k}_a| \left(\sum_b \alpha_{a,g(b)}^2 x_b^2 \right)^{1/2} . \tag{3.56}$$

The ℓ_2 aggregation captures the overall capacity of query dimension a to alter the scalar score through legal unit steps. It is used to partition dimensions for candidate selection; it is not itself the acceptance objective. Every candidate admitted by the sensitivity heuristic must still pass the exact $\Delta\mathcal{L}_{ab} < 0$ test.

Kneedle-Based Protection and Candidate Region

The dimension sensitivities are sorted in descending order:

$$\mathcal{S}_{(1)} \geq \mathcal{S}_{(2)} \geq \cdots \geq \mathcal{S}_{(d_{\text{out}})}. \tag{3.57}$$

Kneedle [1] is applied to this sorted curve to locate a knee index a_{knee} that separates a steep extreme regime from the flatter remainder. Dimensions before the knee,

$$\mathcal{P} = \{(1), \ldots, (a_{\text{knee}})\}, \tag{3.58}$$

are protected from ReFlip. A unit integer move in these dimensions can cause a disproportionately large score change, so excluding them reduces the risk of overshooting the target or making the result overly dependent on a single extreme coordinate.

ReFlip permits candidates in a moderate-sensitivity window immediately after the knee. With a window-width hyperparameter h_{R},

$$\mathcal{M} = \{(a_{\text{knee}} + 1), \ldots, \min(a_{\text{knee}} + h_{\text{R}}, d_{\text{out}})\} . \tag{3.59}$$

The original output-dimension indices associated with these sorted positions define the allowed rows of $\hat{W}_Q$. This region retains enough score influence to support useful correction while avoiding the most extreme dimensions. Within each allowed row, legal parameter candidates are evaluated using $\mathcal{S}_{ab}$ as an ordering signal and $\Delta\mathcal{L}_{ab}$ as the acceptance criterion.

Kneedle supplies an adaptive partition when sensitivity distributions differ across layers. Nevertheless, it is a heuristic. The detected knee can depend on normalization, smoothing, and the Kneedle sensitivity setting; it does not guarantee the globally optimal protected set, allowed set, or final discrete configuration. The post-knee window width is therefore retained as an explicit hyperparameter, and all accepted changes remain governed by the exact scalar loss.

Greedy ReFlip Procedure

ReFlip proceeds from the current $\hat{W}_Q$, fixed $\hat{W}_K$, and exact residual e_s. It first computes $\mathcal{S}_{ab}$ and $\mathcal{S}_a$, sorts the dimension sensitivities, applies Kneedle, protects the pre-knee extreme dimensions, and selects the moderate region immediately after the knee. It then evaluates legal one-step changes to query integers in the allowed rows.

For each candidate $(a, b, \Delta z_{ab})$, the procedure computes

$$\Delta s_{ab} = \hat{k}_a x_b \alpha_{a,g(b)} \Delta z_{ab} \tag{3.60}$$

and

$$\Delta \mathcal{L}_{ab} = e_s \Delta s_{ab} + \frac{1}{2}(\Delta s_{ab})^2. \tag{3.61}$$

Candidates with nonnegative $\Delta \mathcal{L}_{ab}$ are rejected. An accepted candidate updates the integer, $\hat{q}_a$, $\hat{s}$, and e_s before the next evaluation. If both directions are legal and beneficial, the direction with the more negative exact loss change is preferred. The procedure terminates when it reaches its accepted-change budget, exhausts its pass budget, or finds no loss-decreasing candidate.

Because the residual is scalar and updated after each accepted change, the exact condition has a transparent interpretation:

$$|e_s + \Delta s_{ab}| < |e_s|. \tag{3.62}$$

ReFlip accepts a change only when it moves the current quantized score closer to the original scalar target. The quadratic term prevents a large score step from being accepted when it crosses the target by more than it corrects.

Scope and Limitations of ReFlip

ReFlip directly targets a query–key interaction that Flip does not model. In particular, a residual caused partly by $\hat{W}_K - W_K$ can be reduced by a compensating change in $\hat{W}_Q$. This is a functional correction: the selected query weight need not become closer to its original full-precision value. As in Flip, individual reconstruction error may increase while the targeted aggregate objective improves.

The method deliberately avoids a full-matrix Taylor or gradient formulation. Its candidate score change and loss change are exact for the scalar objective defined above. This precision also clarifies the limitation: ReFlip does not directly preserve every query–key score between distinct token rows, the scaled logits, the softmax distribution, or the final value-weighted attention output. A lower scalar loss under the representative activation is evidence only for that targeted criterion.

ReFlip is also greedy and discrete. Candidate ordering and the protected sensitivity region can affect the result, and a sequence of individually rejected changes could conceivably be beneficial if applied jointly. Kneedle does not remove this limitation and does not provide an optimality certificate. The procedure is therefore best understood as a conservative post-training heuristic with exact local acceptance under a clearly bounded objective.

3.7 Unified Refinement Procedure

The complete methodology follows a consistent sequence from initialization to local projection correction and then to scalar interaction correction.

Stage 1: Calibration and Quantization

First, calibration activations are summarized into the empirical statistic $\tilde{x}$ and then shrunk toward target m to obtain the representative activation x. Next, each of W_Q, W_K, and W_V is partitioned along the input dimension into groups. Group-specific asymmetric INT4 scales and zero points are computed, and RTN initializes the stored integers. This stage produces $\hat{W}_Q$, $\hat{W}_K$, and $\hat{W}_V$ under a common legal integer representation.

Stage 2: Flip

For each projection selected for Flip, compute

$$E = \hat{W} - W, \qquad r = xE^\top, \qquad J = \|r\|_2^2. \tag{3.63}$$

Evaluate legal one-step integer candidates. For a candidate at (j, i), use

$$\Delta J = 2r_j x_i \delta + (x_i \delta)^2, \qquad \delta = \alpha_{j,g(i)} \Delta z. \tag{3.64}$$

Accept only a strictly negative exact objective change and immediately update r_j. Repeat until the Flip stopping rule is met. The result is a projection matrix whose aggregate residual under x is no larger than at the beginning of the accepted sequence.

Stage 3: ReFlip

After the chosen Flip refinements, compute

$$s^\star = (xW_Q^\top) \cdot (xW_K^\top), \qquad \hat{s} = (x\hat{W}_Q^\top) \cdot (x\hat{W}_K^\top), \tag{3.65}$$

and set $e_s = \hat{s} - s^\star$. Hold $\hat{W}_K$ fixed for the entire ReFlip stage. Compute parameter and dimension sensitivities from the current $\hat{k}$, apply Kneedle to the sorted $\mathcal{S}_a$ curve, protect the pre-knee extreme dimensions, and admit the moderate region immediately after the knee.

For every allowed query candidate, evaluate

$$\Delta s_{ab} = \hat{k}_a x_b \alpha_{a,g(b)} \Delta z_{ab} \tag{3.66}$$

and

$$\Delta \mathcal{L}_{ab} = e_s \Delta s_{ab} + \frac{1}{2}(\Delta s_{ab})^2. \tag{3.67}$$

Accept only $\Delta \mathcal{L}_{ab} < 0$, update the scalar residual after each accepted change, and terminate under the ReFlip stopping rule. No key or value integer is modified during this stage.

Monotonicity of Accepted Changes

The two stages have simple monotonicity properties under their own stated objectives. Every accepted Flip change satisfies

$$J_{\text{after}} < J_{\text{before}}, \tag{3.68}$$

and every accepted ReFlip change satisfies

$$\mathcal{L}_{\text{after}} < \mathcal{L}_{\text{before}}. \tag{3.69}$$

These properties follow from exact objective differences and immediate residual updates. They do not imply monotonic improvement in a different metric, such as unweighted weight reconstruction error, the complete attention matrix, language-model loss, or downstream task quality.

The order of the stages matters conceptually. Flip first improves each selected projection under an activation-weighted output objective. ReFlip then observes the score residual left by the current quantized query and key matrices and uses query-only compensation. Reversing the stages would change the scalar residual and sensitivity profile against which ReFlip operates.

3.8 Hyperparameters and Design Choices

The methodology exposes its principal choices as explicit hyperparameters.

- **Bit width b.** The present formulation fixes $b = 4$, giving the legal unsigned integer set $\mathcal{Z}_4 = \{0, \ldots, 15\}$.

- **Group size** g**.** The group size determines the mapping $g(b)$ and the granularity of scales and zero points. Smaller groups increase quantization metadata but can better adapt to local weight ranges.
- **Calibration sample count** N_{cal}**.** This controls the amount of activation data used to estimate $\tilde{x}$, the shrinkage target, and the noise level.
- **James–Stein target** m **and noise estimate** $\hat{\sigma}^2$**.** These determine the shrinkage factor λ and therefore the representative activation used by both exact objectives.
- **Flip pass budget** T_{F}**.** This is the maximum number of scans over eligible Flip coordinates.
- **Flip acceptance budget** B_{F}**.** This limits the total number of accepted integer changes for a projection or layer.
- **Flip improvement tolerance** ε_{F}**.** In finite precision, the strict mathematical condition $\Delta J < 0$ may be implemented conservatively as $\Delta J < -\varepsilon_{\text{F}}$.
- **ReFlip Kneedle sensitivity** κ**.** This controls how the knee detector responds to curvature in the sorted dimension-sensitivity curve.
- **Moderate-region width** h_{R}**.** This determines how many sorted dimensions immediately after the knee are eligible for ReFlip.
- **ReFlip pass budget** T_{R} **and acceptance budget** B_{R}**.** These bound the greedy search over legal query-integer changes.
- **ReFlip improvement tolerance** ε_{R}**.** A conservative finite-precision rule accepts only $\Delta\mathcal{L}_{ab} < -\varepsilon_{\text{R}}$.
- **Candidate order.** Flip candidates may be ordered by potential activation-weighted step magnitude, while ReFlip candidates may be ordered by $\mathcal{S}_{ab}$. Ordering affects the greedy path but never replaces the exact acceptance tests.

The one-step neighborhood $\Delta z \in \{-1, +1\}$ is part of the method rather than an unconstrained hyperparameter. It keeps changes local on the existing INT4 grid and makes both objective differences exact and inexpensive to evaluate. Multiple accepted one-step changes may eventually move an integer by more than one level if the candidate is revisited and every intermediate change remains legal and strictly beneficial.

Hyperparameter selection should respect the distinction between refinement opportunity and refinement risk. Very large budgets or a wide post-knee region create more opportunities to reduce the calibration objective but can increase dependence on the representative activation. Conservative budgets, shrinkage, protected extreme dimensions, and positive improvement tolerances provide regularization against such over-specialization.

3.9 Theoretical Discussion

From Scalar Reconstruction to Functional Residuals

The methodology is motivated by a hierarchy of objectives. RTN minimizes isolated scalar weight error for fixed quantization parameters. Flip moves to a projection-level functional residual,

$$x(\hat{W} - W)^\top, \tag{3.70}$$

which accounts for activation magnitude, error sign, and cancellation within each output row. ReFlip then moves to a scalar interaction objective,

$$\frac{1}{2}\left[(x\hat{W}_Q^\top) \cdot (x\hat{W}_K^\top) - (xW_Q^\top) \cdot (xW_K^\top)\right]^2, \tag{3.71}$$

which accounts for the coupling between query and key projections while constraining changes to the quantized query matrix.

Each level deliberately sacrifices breadth for tractability. The objectives are compact enough to permit exact evaluation of every legal one-step candidate. This avoids the ambiguity of claiming that a local scalar change optimizes a large downstream object through an unverified approximation. The trade-off is that exactness applies only to the specified representative objectives.

Why Query-Only Compensation Can Help

The score residual can be expanded conceptually as the difference between a dot product of two quantized projected vectors and the corresponding full-precision dot product. Its value reflects errors in both vectors and their interaction. Once $\hat{W}_K$ is fixed, changing $\hat{W}_Q$ alters one side of the dot product while leaving the other side as the reference direction for correction. A suitable query change can therefore offset part of the score error associated with the fixed quantized key.

This compensation does not require the modified query weight to approach its original value. It requires only that the exact change in $\hat{s}$ reduce $|\hat{s} - s^\star|$. ReFlip is thus aligned with the same general principle as Flip: a functional aggregate can improve even when an individual scalar reconstruction becomes worse.

Role of Sensitivity Protection

Sensitivity has two competing interpretations. A high-sensitivity parameter can correct a residual with a small number of integer changes, but it can also overshoot the target or make the result fragile to calibration variation. ReFlip therefore does not simply choose the largest sensitivities. It protects dimensions in the extreme pre-knee region and searches the moderate region immediately after the knee.

This policy is intentionally risk-aware rather than optimal. The Kneedle partition adapts to the shape of each layer's sensitivity distribution, while the exact loss test ensures that an admitted candidate is accepted only when it improves the current scalar objective. The separation between heuristic admission and exact acceptance is central: sensitivity controls where to search, whereas $\Delta\mathcal{L}_{ab}$ determines whether to change an integer.

Boundaries of the Claims

Several boundaries follow directly from the formulation. First, the representative activation is a James–Stein-shrunk calibration summary, not the complete activation distribution. Second, Flip exactly reduces only its projection residual objective for accepted changes. Third, ReFlip exactly reduces only its representative scalar query–key score loss for accepted changes. Fourth, ReFlip modifies $\hat{W}_Q$ only and makes no update to $\hat{W}_K$ or $\hat{W}_V$.

Most importantly, the ReFlip scalar objective does not preserve the full pairwise attention matrix. In a complete sequence, queries and keys arise from different token rows, and attention logits include many cross-token dot products. A single score based on x cannot constrain all of them. The method also does not directly optimize the softmax transformation or the value-weighted attention output. These limitations do not invalidate the targeted objectives, but they prevent stronger claims than the mathematics supports.

Discrete Search and Optimality

Both Flip and ReFlip perform greedy coordinate search on a finite integer grid. Their accepted changes are monotonic under their respective objectives, and termination is guaranteed under finite budgets. Without an explicit budget, strict decrease on a finite state space also rules out cycles, although repeated scans may be undesirable.

Neither procedure guarantees a globally optimal quantized matrix. Flip can miss coordinated changes that are beneficial only when applied together. ReFlip can miss a sequence whose first step is neutral or unfavorable but whose combined effect is favorable. Candidate ordering, budget choices, and the Kneedle-selected region can all influence the endpoint. Kneedle itself is a heuristic detector, not an optimality guarantee. The methodological claim is therefore limited to exact evaluation and strict improvement of accepted local changes under the stated objectives.

3.10 Summary

This chapter has defined a post-training refinement methodology for group-wise asymmetric INT4 attention projections under a consistent row-vector convention. A representative James–Stein-shrunk activation x provides the calibration signal. The initial RTN quantizer assigns legal integers z_{ab} with group-specific scales $\alpha_{a,g(b)}$ and zero points $\zeta_{a,g(b)}$.

Flip refines a projection matrix through the exact objective

$$J = \left\| x(\hat{W} - W)^\top \right\|_2^2 . \tag{3.72}$$

For a legal dequantized change δ at (j, i), it evaluates

$$\Delta J = 2r_j x_i \delta + (x_i \delta)^2 \tag{3.73}$$

and accepts only strict reductions. RTN remains the minimizer of isolated scalar weight error; Flip can improve the aggregate activation-weighted residual by allowing signed error cancellation, even when an individual weight becomes less accurate.

ReFlip holds $\hat{W}_K$ fixed and updates $\hat{W}_Q$ only. It targets

$$\mathcal{L} = \frac{1}{2}\left[(x\hat{W}_Q^\top) \cdot (x\hat{W}_K^\top) - (xW_Q^\top) \cdot (xW_K^\top)\right]^2 . \tag{3.74}$$

A legal unit integer change has exact score effect

$$\Delta s_{ab} = \hat{k}_a x_b \alpha_{a,g(b)} \Delta z_{ab}, \tag{3.75}$$

parameter sensitivity

$$\mathcal{S}_{ab} = |\hat{k}_a x_b \alpha_{a,g(b)}|, \tag{3.76}$$

and dimension sensitivity

$$\mathcal{S}_a = |\hat{k}_a| \left(\sum_b \alpha_{a,g(b)}^2 x_b^2 \right)^{1/2} . \tag{3.77}$$

Kneedle protects pre-knee extreme dimensions and admits a moderate region immediately after the knee. Each admitted candidate is evaluated using the exact loss change

$$\Delta\mathcal{L}_{ab} = (\hat{s} - s^{\star})\Delta s_{ab} + \frac{1}{2}(\Delta s_{ab})^2, \tag{3.78}$$

and the scalar residual is updated after every accepted change.

The resulting procedure retains the theoretical memory advantages and deployment form of group-wise INT4 weight-only quantization. Its guarantees are deliberately limited to the stated objectives. Kneedle is heuristic, greedy refinement is not globally optimal, and the scalar ReFlip objective does not preserve the full pairwise attention matrix. Within these boundaries, Flip first corrects the projection residual under the representative activation, and ReFlip then performs exact compensation of the scalar query–key score.

4.1 Experimental Setup

This chapter evaluates whether the proposed Flip and ReFlip refinements improve quantization quality beyond standard round-to-nearest quantization and established post-training quantization baselines.

Two levels of evaluation are considered. The first is a standalone QKV experiment, where a selected grouped-query-attention block is isolated and the attention-score distortion is measured directly. This experiment makes the local effect of each refinement visible before running the full model. The second is a full-model evaluation, where quantized models are compared using perplexity and downstream task accuracy.

The standalone experiment follows the Transformer attention formulation [12] and focuses on the grouped-query-attention (GQA) structure used by modern decoder-only language models [14]. It uses Llama-3-8B, layer 8, GQA group 3. Calibration activations are collected from C4 [16] using 128 samples with sequence length 512, producing an activation matrix of size 65536×4096. The exported group contains four query heads sharing one key/value group. James-Stein shrinkage [2] is used to stabilize the activation mean estimate before quantization decisions are evaluated.

The full-model comparison covers Mistral-7B [17], Llama-3/Llama-3.1 8B [18], and Qwen2.5-7B [19]. The results are taken from the 4-bit sheet of the reference workbook. Only the metrics used in this thesis are retained: WikiText-2 perplexity, C4 perplexity, ARC-Challenge, ARC-Easy, BoolQ, PIQA, RTE, and the average of the five accuracy tasks. WikiText-2 is a language modeling benchmark introduced with the WikiText corpus [20], while C4 originates from the T5 work [16]. ARC-Challenge and ARC-Easy measure science-question reasoning [21]; BoolQ measures natural yes/no question answering [22]; PIQA measures physical commonsense reasoning [23]; and RTE is included through the GLUE benchmark family [24]. The lm-evaluation harness provides a reproducible framework for running these tasks consistently across language models [25].

For WikiText-2 and C4, lower perplexity is better. For ARC-Challenge, ARC-Easy, BoolQ, PIQA, RTE, and their average, higher accuracy is better. The reported average is computed over the five selected downstream accuracy metrics:

$$\text{avg} = \frac{\text{arc-c} + \text{arc-e} + \text{boolq} + \text{piqa} + \text{rte}}{5}.$$

Full precision is shown as a reference ceiling rather than as a competitor in the quantized-method ranking. Therefore, in the full-model tables, bold values indicate the best quantized result for a metric within the same model, and underlined values indicate the second-best quantized result. Full-precision values are not highlighted.

The method names are normalized for clarity. The full-precision model is denoted as *Full precision*. Standard round-to-nearest quantization is denoted as *RTN*. RTN represents the direct rounding baseline commonly used in post-training quantization before more data-aware reconstruction is applied [5], [26], [27]. GPTQ uses approximate second-order information to quantize pretrained Transformer weights [7]. AWQ is an activation-aware weight-only quantization method that protects salient channels using calibration statistics [8].

4.2 Standalone QKV Evaluation

The standalone QKV experiment compares three strategies under the same INT4 group-wise quantization setting. The nearest strategy performs direct round-to-nearest quantization. Flip then adjusts selected integer values by one quantization level to reduce local reconstruction error while avoiding activation outlier channels. ReFlip performs a second-stage correction that targets the attention-score error remaining after Flip.

Table 4.1: Standalone QKV attention-score error on Llama-3-8B layer 8, GQA group 3.

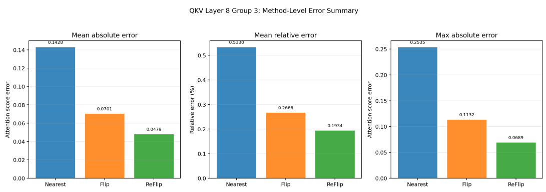
Method	Mean abs. error	Max abs. error	Mean rel. error (%)	Reduction vs. RTN (%)	Integer flips
RTN	0.142806	0.253523	0.5330	–	–
Flip	0.070070	0.113184	0.2666	41.63	20,889
ReFlip	**0.047939**	**0.068937**	**0.1934**	**56.63**	64

Table 4.1 shows that Flip substantially reduces the attention-score error compared with RTN, while ReFlip provides an additional reduction with very few extra integer changes. Flip modifies 20,889 weights across the selected Q and K projections, corresponding to approximately 0.7969% of the involved weights. ReFlip applies only 64 additional integer changes, but still reduces the mean absolute attention-score error from 0.070070 to 0.047939.

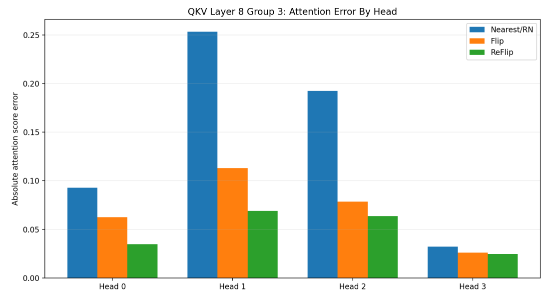
Table 4.2: Per-head attention-score errors in the standalone QKV experiment.

Head	RTN error	Flip error	ReFlip error	RTN to Flip (%)	RTN to ReFlip (%)
0	0.092831	0.062511	**0.034649**	32.66	62.68
1	0.253523	0.113184	**0.068937**	55.36	72.81
2	0.192633	0.078560	**0.063698**	59.22	66.93
3	0.032239	0.026023	**0.024472**	19.28	24.09

The per-head results in Table 4.2 indicate that ReFlip improves all four query heads in the selected GQA group. The magnitude of the improvement is not uniform across heads: head 1 obtains the largest RTN-to-ReFlip reduction, while head 3 shows a smaller but still positive reduction. This difference is expected because each query head has a different original score magnitude, error direction, and set of useful moderate dimensions for correction.

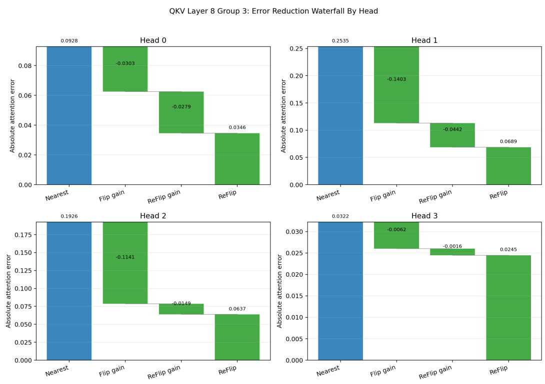


(a) Mean and maximum attention errors.

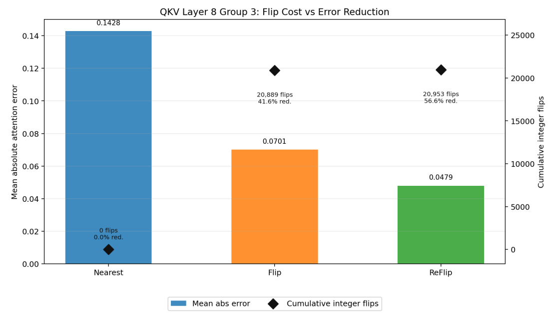


(b) Attention error by query head.

Figure 4.1: Standalone QKV error comparison across RTN, Flip, and ReFlip.



(a) Error reduction from RTN to ReFlip.



(b) Flip cost versus error reduction.

Figure 4.2: Per-head correction behavior and cost-benefit analysis.

The plots in Figures 4.1 and 4.2 support the numerical tables. Flip provides a large first correction by changing a small fraction of weights, while ReFlip targets the remaining attention-score error more directly. The ReFlip stage therefore behaves as a fine correction step rather than a broad re-quantization pass.

4.3 Full-Model Evaluation

The full-model results evaluate whether the local improvements observed in the QKV setting are reflected in end-to-end model behavior. Four model families are included: Mistral-7B, Llama-3.1-8B, Qwen2.5-7B, and Llama-3-8B. For each model, the tables report 4-bit quantization results for RTN, AWQ, GPTQ, and their Flip/ReFlip variants when the corresponding workbook entries are available.

Table 4.3: Full-model 4-bit evaluation for Mistral-7B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	4.845	7.535	0.509	0.802	0.837	0.800	0.686	0.7268
RTN	4.9917	7.8047	0.4761	0.7875	<u>0.8248</u>	0.7992	0.6643	0.7104
RTN + ReFlip	4.9704	<u>7.7167</u>	0.4804	0.7934	0.8202	<u>0.8014</u>	0.6751	0.7141
RTN + Flip	4.9948	7.738	0.4744	0.7887	**0.8275**	0.7987	0.6606	0.7100
AWQ	4.975	7.727	<u>0.493</u>	0.794	0.817	0.797	0.671	0.7144
AWQ + ReFlip	**4.965**	**7.711**	**0.495**	<u>0.795</u>	0.818	**0.802**	**0.690**	**0.7200**
AWQ + Flip	4.975	7.726	<u>0.493</u>	0.793	0.815	0.798	<u>0.678</u>	0.7154
GPTQ	4.978	7.744	0.480	<u>0.795</u>	0.820	0.798	0.675	0.7136
GPTQ + ReFlip	<u>4.967</u>	7.736	0.483	**0.796**	0.821	0.800	<u>0.678</u>	<u>0.7156</u>
GPTQ + Flip	4.979	7.745	0.480	<u>0.795</u>	0.819	0.798	0.675	0.7134

Table 4.4: Full-model 4-bit evaluation for Llama-3.1-8B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	5.522	8.409	0.522	0.822	0.830	0.791	0.711	0.7352
RTN	6.0248	9.4397	0.4898	0.8064	<u>0.7990</u>	0.7835	0.6351	0.7147
RTN + ReFlip	5.9727	9.3692	0.5051	<u>0.8068</u>	0.7928	0.7867	0.6531	**0.7209**
RTN + Flip	6.0815	9.6308	0.4872	0.8009	**0.8075**	0.7845	0.6315	0.7143
AWQ	<u>5.968</u>	9.321	<u>0.511</u>	0.806	0.794	**0.798**	0.635	0.7088
AWQ + ReFlip	**5.949**	**9.294**	**0.512**	**0.808**	0.796	0.796	**0.671**	<u>0.7166</u>
AWQ + Flip	<u>5.968</u>	9.316	0.507	0.803	0.790	0.796	0.632	0.7056
GPTQ	5.995	9.304	0.505	0.804	0.795	0.793	0.651	0.7096
GPTQ + ReFlip	5.989	<u>9.300</u>	0.507	0.806	0.790	<u>0.797</u>	<u>0.663</u>	0.7126
GPTQ + Flip	5.992	9.304	0.499	0.803	0.792	0.793	0.659	0.7092

Table 4.5: Full-model 4-bit evaluation for Qwen2.5-7B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	6.0839	10.8166	0.477	0.804	0.846	0.787	0.815	0.7458
RTN	6.6915	12.2029	0.4506	0.7784	0.8302	0.7662	0.8016	0.7254
RTN + ReFlip	6.6366	11.8779	0.45162	0.77125	0.83264	0.76782	0.816018	0.7278696
RTN + Flip	6.6427	12.1076	0.44138	0.76832	0.82685	0.7651	**0.823258**	0.7249816
AWQ	6.4677	<u>11.4443</u>	<u>0.469</u>	0.786	0.842	0.780	0.805	0.7364
AWQ + ReFlip	**6.438**	**11.363**	**0.473**	**0.792**	**0.847**	<u>0.782</u>	<u>0.819</u>	**0.7426**
AWQ + Flip	<u>6.467</u>	11.450	0.459	<u>0.787</u>	<u>0.843</u>	**0.787**	0.815	<u>0.7382</u>
GPTQ	6.708	11.539	0.454	0.781	0.805	0.780	0.769	0.7178
GPTQ + ReFlip	6.702	11.534	0.459	0.779	0.814	0.779	0.787	0.7236
GPTQ + Flip	6.709	11.538	0.451	0.779	0.805	0.778	0.769	0.7164

Table 4.6: Full-model 4-bit evaluation for Llama-3-8B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	5.442	8.346	0.505	0.800	0.813	0.796	0.696	0.7220
RTN	6.0572	9.5846	0.4789	0.780	0.791	0.784	0.628	0.6924
RTN + ReFlip	6.0236	9.2523	<u>0.490</u>	0.781	0.796	0.7818	0.6679	0.7033
RTN + Flip	6.0711	9.3279	0.4806	0.782	0.794	0.7802	0.6390	0.6952
AWQ	5.910	9.240	0.488	0.784	0.795	<u>0.791</u>	0.646	0.7008
AWQ + ReFlip	**5.885**	**9.199**	**0.493**	<u>0.788</u>	**0.799**	0.788	0.679	<u>0.7094</u>
AWQ + Flip	<u>5.906</u>	<u>9.234</u>	0.489	0.784	0.796	**0.792**	0.617	0.6956
GPTQ	6.011	9.370	0.487	0.783	<u>0.798</u>	0.788	<u>0.689</u>	0.7090
GPTQ + ReFlip	6.013	9.354	0.489	**0.790**	0.795	0.788	**0.693**	**0.7110**
GPTQ + Flip	6.014	9.371	0.486	0.783	0.795	0.787	0.686	0.7074

4.4 Discussion

The standalone QKV experiment gives the clearest evidence for the mechanism targeted by this thesis. RTN introduces attention-score distortion because independent rounding errors in Q and K interact through the dot product. Flip reduces this error by improving local rounding decisions on non-outlier channels, and ReFlip further reduces the remaining error by explicitly correcting attention-score mismatch. The reduction from 0.142806 to 0.047939 in mean absolute attention error shows that the two-stage refinement is effective in the isolated setting. The cost profile is also important: Flip performs a broader local correction, while ReFlip adds only 64 integer flips and acts as a targeted residual correction step.

The full-model results are more nuanced because the downstream tasks depend on the entire network, not only on one QKV group. Nevertheless, several consistent patterns appear. On Mistral-7B, AWQ + ReFlip is the strongest quantized method for WikiText-2, C4, ARC-Challenge, PIQA, RTE, and the selected-task average. This suggests that the activation-aware scaling of AWQ and the attention-aware residual correction of ReFlip are complementary in this case. GPTQ + ReFlip obtains the best quantized ARC-Easy score, while RTN + Flip obtains the best quantized BoolQ score, showing that the best refinement can still vary by task.

On Llama-3.1-8B, AWQ + ReFlip gives the best quantized perplexity and the best quantized scores on ARC-Challenge, ARC-Easy, and RTE. RTN + ReFlip gives the highest quantized average, improving the RTN average from 0.7147 to 0.7209 while also reducing WikiText-2 and C4 perplexity. This indicates that ReFlip can be beneficial even when applied to the simplest RTN baseline, although the improvement is not uniformly distributed across every individual task.

On Qwen2.5-7B, AWQ + ReFlip is the strongest quantized method for WikiText-2, C4, ARC-Challenge, ARC-Easy, BoolQ, and the selected-task average. The RTN + ReFlip row also improves RTN perplexity and average task score, increasing the average from 0.7254 to 0.7279. RTN + Flip is particularly strong on RTE, reaching the best quantized RTE score in the table, but its average is slightly lower than RTN + ReFlip. This supports the interpretation that Flip and ReFlip correct different aspects of quantization error: Flip may strongly benefit selected tasks, while ReFlip tends to provide a more attention-aligned residual correction.

On Llama-3-8B, AWQ + ReFlip gives the best quantized WikiText-2 and C4 perplexity and also leads on ARC-Challenge and BoolQ. GPTQ + ReFlip gives the best quantized ARC-Easy, RTE, and average score. The improvements over each raw baseline are visible: RTN + ReFlip improves the RTN average from 0.6924 to 0.7033, AWQ + ReFlip improves AWQ from 0.7008 to 0.7094, and GPTQ + ReFlip improves GPTQ from 0.7090 to 0.7110. This is the clearest full-model evidence that ReFlip can improve multiple baseline quantizers rather than only a single quantization family.

Overall, ReFlip most consistently improves perplexity and average downstream performance when paired with AWQ or RTN, while GPTQ already provides a stronger reconstruction baseline and therefore leaves a smaller margin for further correction. Flip remains useful but is more task-dependent. This behavior is consistent with the design of the methods: Flip is a local rounding correction, whereas ReFlip explicitly targets the bilinear attention-score mismatch that arises from simultaneous Q and K quantization. The results should therefore be interpreted as evidence that attention-aware correction improves quantized model fidelity, not as a guarantee that every task score must improve after every refinement.

CHAPTER 5. CONCLUSIONS

5.1 Summary

This thesis studied post-training weight quantization for large language models, with a particular focus on the query, key, and value projection matrices inside grouped-query attention. The central motivation is that low-bit quantization, especially INT4 quantization, can greatly reduce model storage and deployment cost, but it can also introduce non-negligible errors in attention computation. Since attention scores are formed from the interaction between quantized query and key projections, their rounding errors may interact and produce larger distortions in the final dot product.

To address this problem, the thesis developed and evaluated two refinement mechanisms, Flip and ReFlip. Flip revisits local rounding decisions after standard round-to-nearest quantization and selectively changes integer quantization levels when doing so reduces the aggregate activation-weighted projection residual. A beneficial Flip may therefore increase the isolated error of an individual weight while improving the projection response through cancellation among weighted errors. ReFlip then performs a second correction stage that holds the quantized key projection fixed and updates only the quantized query projection to reduce the remaining scalar attention-score mismatch. The method is supported by calibration statistics, James–Stein mean estimation, and Kneedle-based dimension-sensitivity selection, which together provide a data-aware way to decide where quantization errors should be corrected and where weights should be left untouched.

The experimental chapter evaluated the proposed refinements at two levels. First, a standalone QKV experiment isolated one Llama-3-8B attention group and measured attention-score distortion directly. Second, full-model results compared RTN, AWQ, GPTQ, and their Flip/ReFlip variants on perplexity and downstream language understanding metrics. This combination of local and end-to-end evaluation gives a more complete view of how attention-aware correction behaves in practice.

5.2 Main Findings

The standalone QKV experiment provides the clearest evidence for the mechanism targeted by this work. On Llama-3-8B layer 8, GQA group 3, standard RTN quantization produced a mean absolute attention-score error of 0.142806. Flip reduced this value to 0.070070, and ReFlip further reduced it to 0.047939. Overall, ReFlip achieved a 56.63% mean error reduction relative to RTN in this isolated setting. This confirms that refining the quantized query projection with scalar attention-score feedback can reduce the specific distortion that the method is designed to address.

The cost of the correction is also important. Flip changed 20,889 integer weights in the selected Q and K projections, corresponding to less than one percent of the involved weights. ReFlip then added only 64 integer changes while still producing an additional error reduction. This suggests that ReFlip acts as a targeted residual correction step rather than a broad re-quantization procedure.

The full-model results show that the local QKV improvement can translate into better model-level behavior, although the effect is task-dependent. Across the evaluated models, AWQ + ReFlip is often the strongest quantized method for perplexity and average downstream performance. This is especially visible on Mistral-7B, Qwen2.5-7B, and Llama-3-8B, where AWQ + ReFlip gives strong WikiText-2 and C4 perplexity while also improving several downstream metrics. For Llama-3-8B, GPTQ + ReFlip gives the highest selected-task average among quantized methods, showing that ReFlip can also complement a second-order reconstruction baseline.

The results also show that Flip and ReFlip do not behave identically. Flip can produce strong gains on selected tasks, such as the RTE result observed for Qwen2.5-7B under RTN + Flip, while ReFlip tends to provide a more consistent correction of residual attention-score error. This supports the interpretation that Flip primarily refines the aggregate activation-weighted projection residual, whereas ReFlip is more directly aligned with the bilinear structure of attention scores.

5.3 Limitations

The selected metric set is limited. WikiText-2 and C4 evaluate language modeling perplexity, while ARC-Challenge, ARC-Easy, BoolQ, PIQA, and RTE evaluate downstream accuracy. These metrics are useful, but they do not cover every type of reasoning, long-context behavior, generation quality, latency, memory bandwidth, or hardware-level inference efficiency. A complete deployment study would need to include these additional dimensions.

5.4 Future Work

Firstly, future experiments should evaluate a larger task suite and more deployment-relevant metrics. Additional lm-evaluation tasks, long-context benchmarks, runtime measurements, memory usage, and throughput would clarify whether the quality gains from ReFlip also lead to practical advantages during inference.

Secondly, the correction strategy itself can be improved. ReFlip uses exact scalar-score loss changes within a candidate region selected heuristically from the sensitivity distribution, and it modifies only the query projection while holding the key projection fixed. Future work may explore stronger optimization objectives, layer-adaptive flip budgets, second-order approximations for Q–K interactions, and hardware-aware constraints so that the method can balance accuracy, efficiency, and implementation cost more effectively.

An additional direction is to extend ReFlip to the value projection using a loss defined on the full attention output rather than on the attention score. For fixed query and key projections, let

$$A = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right), \qquad O = AV = AXW_V^\top.$$

Since W_V does not affect the attention logits or the attention probabilities, a value-projection correction cannot be justified by the scalar Q–K score used in the present ReFlip formulation. It can instead be evaluated through the residual of O, with A held fixed during each W_V-only update. This extension would connect discrete value-weight refinement directly to the representation produced by the complete attention operation.

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